

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ALCON INC., ALCON VISION, LLC, ALCON LABORATORIES, INC.,
and ALCON RESEARCH, LLC.,
Petitioner,

v.

AMO DEVELOPMENT, LLC,
Patent Owner.

IPR2021-00845
Patent 9,233,024 B2

Before SHERIDAN K. SNEDDEN, JON B. TORNQUIST, and
RYAN H. FLAX, *Administrative Patent Judges*.

TORNQUIST, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. *Background and Summary*

Alcon Inc., Alcon LenSx, Inc., Alcon Vision, LLC, Alcon Laboratories, Inc., and Alcon Research, LLC (collectively “Petitioner”) filed a Petition (Paper 1, “Pet.”) requesting an *inter partes* review of claims 1–17 and 20–26 of U.S. Patent No. 9,233,024 B2 (Ex. 1007, “the ’024 patent”). AMO Development, LLC (“Patent Owner”) filed a Preliminary Response to the Petition. Petitioner then filed a reply to Patent Owner’s Preliminary Response (Paper 11, “Preliminary Reply”) addressing whether Weikert is prior art to the ’024 patent, to which Patent Owner filed a sur-reply (Paper 13).

Upon review of the parties’ arguments and supporting evidence, we instituted review. Paper 15 (“Decision” or “Dec.”).

Patent Owner subsequently filed a Response (Paper 31, “PO Resp.”), to which Petitioner filed a Reply (Paper 39, “Pet. Reply”), and Patent Owner filed a Sur-Reply (Paper 49, “Sur-Reply”).

Petitioner relies, *inter alia*, upon the declaration and reply declaration of Holger Lubatschowski, Ph.D. (Exs. 1001, 1069) and the declaration of Richard Tipperman, M.D. (Ex. 1070). Patent Owner submits the declarations of Jin U. Kang, Ph.D. (Exs. 2002, 2062) and Kathryn M. Hatch, M.D. (Ex. 2004, 2063).

An oral hearing was held on August 30, 2022, and a transcript of the hearing is included in the record (Paper 53, “Tr.”).

For the reasons that follow, we conclude that Petitioner has proven by a preponderance of the evidence that claims 1–17 and 20–26 of the ’024 patent are unpatentable.

B. Real Parties-in-Interest

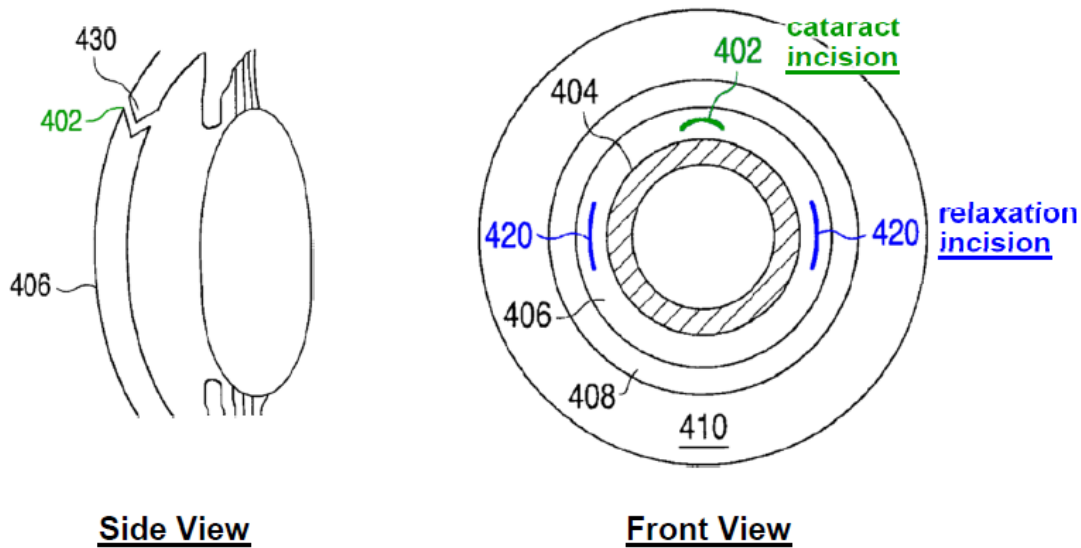
Petitioner identifies Alcon Inc., Alcon Vision, LLC, Alcon Laboratories, Inc., and Alcon Research, LLC as the real parties-in-interest, noting that after the Petition was filed “Alcon LenSx, Inc. merged into Alcon Research, LLC, with Alcon Research LLC the surviving entity.” Paper 3, 1; Pet. 2–3. Patent Owner identifies itself and Johnson & Johnson Surgical Vision, Inc., AMO Manufacturing USA, LLC, and AMO Sales and Services, Inc., as the real parties-in-interest. Paper 5, 1.

C. Related Matters

The '024 patent is asserted in *AMO Development, LLC et al. v. Alcon LenSx, Inc. et al.*, No. 1:20-cv-00842-CFC (D. Del). Pet. 2; Paper 5, 1. *Inter partes* review petitions were also filed by Petitioner against related patents in IPR2021-00843, -00846, and -00849. Paper 3, 1; Pet. 2–3.

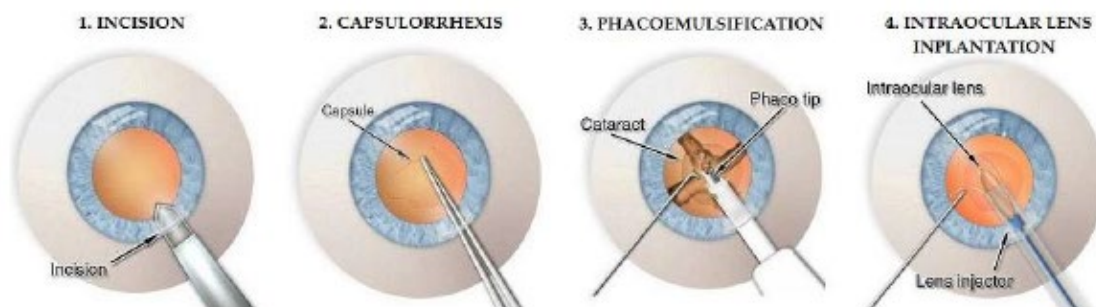
D. Background of Cataract Surgery

Each independent claim of the '024 patent is directed to a “cataract surgery method of treating target tissue in one or more of a cornea, limbus, or sclera of a patient’s eye.” Ex. 1007, 14:7–8, 14:49–50, 15:15–16, 16:20–21. For context, Patent Owner provides annotated figures of the '024 patent, colored to highlight surgical incisions, which we reproduce below (PO Resp. 8):



The figures above are annotated versions of the '024 patent's Figures 7 (above-right) and 8 (above-left) and show, respectively, a front and cross-sectioned side view of the eye's anatomy. Ex. 1007, 2:42–45, Figs. 7, 8. The annotated figures show iris 404, cornea 406, limbus 408, and sclera 410, with cataract incision 402 annotated in green and relaxation incisions 420 annotated in blue (Figure 7). *Id.* at 12:21–26; PO Resp. 8–9.

The parties explain that cataract surgery typically involves removal of the eye's natural lens and replacing it with an intraocular lens (IOL). Ex. 1001 ¶ 23; Ex. 2063 ¶ 26. Petitioner's declarant, Dr. Lubatschowski, provides the following figures depicting these implantation steps (Ex. 1001 ¶ 23):



The figures above show the steps used in the process of implanting an IOL in a patient. *Id.* First, to access the lens, an incision must be made in the cornea or nearby tissues. *Id.* Then an optional step may be implemented to correct for astigmatism, “whether pre-existing or surgery-induced.” *Id.*

¶¶ 23–24. Second, in a process called capsulorhexis or anterior capsulotomy, an opening is made in the anterior lens capsule. *Id.* Third, the eye’s lens is broken apart and removed, usually by ultrasonic phacoemulsification or segmenting the lens cortex and nucleus. *Id.*; Ex. 2063 ¶ 26. Finally, an IOL is implanted in the lens capsule. Ex. 1001 ¶ 23; Ex. 2063 ¶ 26.

E. The ’024 Patent

As noted above, the ’024 patent is directed to cataract surgery methods for treating target tissue in one or more of the cornea, limbus, or sclera of a patient’s eye. Ex. 1007, 1:15–16, 2:13–21, 14:7–8. According to the ’024 patent, the techniques and systems disclosed provide for “rapid and precise openings in the cornea and/or limbus,” and the accuracy and precision by which these incisions are made is “improved over traditional methods.” *Id.* at 2:52–57.

The system of the ’024 patent may be used to create a “cataract incision,” which allows access for the lens removal instrumentation used during cataract surgery. *Id.* at 10:19–25. The ’024 patent explains that a complete cataract incision is not desirable in all situations, such as when the incision is made in an unsterile field where opening the eye to the environment poses further risks of endophthalmitis. *Id.* at 10:29–32. “In this case, the present invention may provide a cataract incision that only partially penetrates” the cornea, limbus, or sclera. *Id.* at 10:32–34.

The '024 patent states that “[m]any cataract patients are astigmatic.” *Id.* at 1:39. One use of the disclosed method is to “perform astigmatic keratotomy, such as limbal and corneal relaxing incisions, in conjunction with the creation of [a] surgical incision that provides the surgeon access to the anterior chamber of an eye.” *Id.* at 2:59–63. To assist in this process, a “wavefront sensor, interferometer, surface profiler, or other such device may be used to yield prescriptions for correcting the astigmatism or other visual aberrations.” *Id.* at 2:65–3:1.

Figure 6 of the '024 patent is reproduced below:

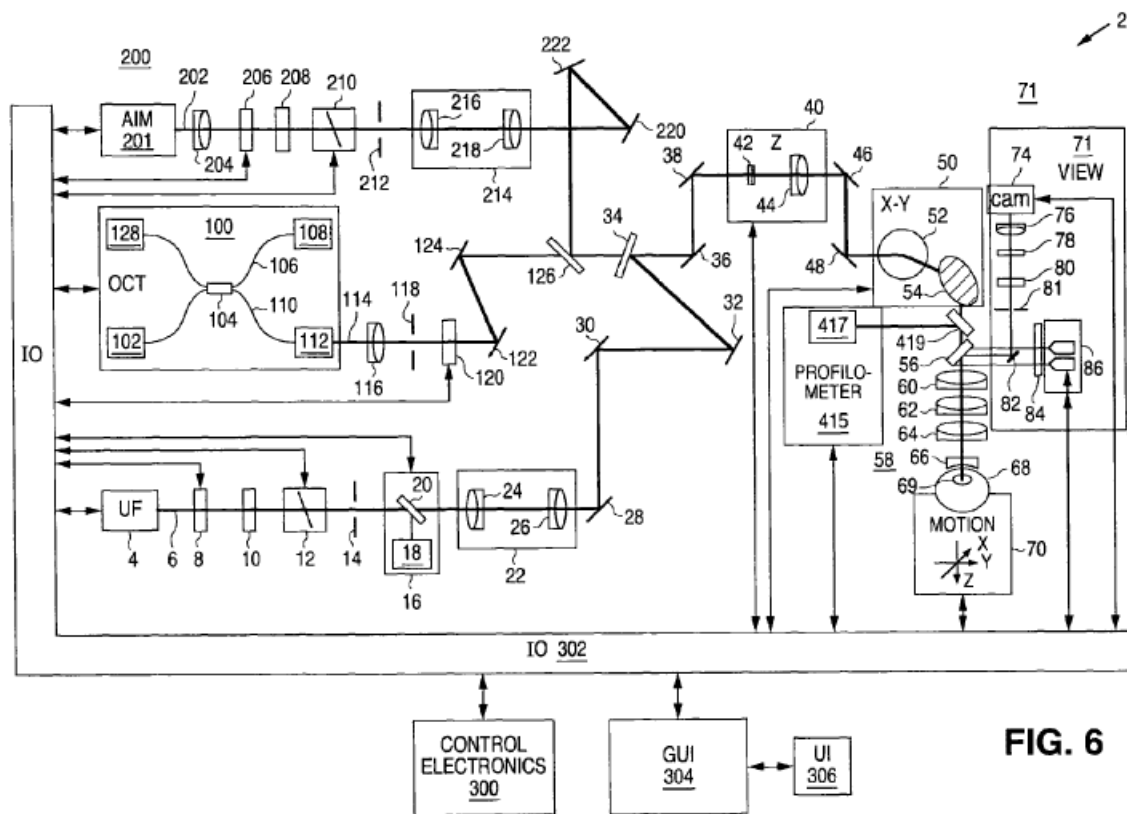


FIG. 6

Figure 6 is a schematic diagram of the optical beam scanning system with a profilometer subsystem. *Id.* at 2:40–41. In the system depicted in Figure 6, ultrafast light source 4 (e.g., a femtosecond laser) is scanned in patient's eye 68 in three dimensions. *Id.* at 3:45–50. Laser 4 is controlled by control electronics 300, via an input and output device 302, to create optical beam 6.

Id. at 4:1–3. “[G]raphical user interface GUI 304 may be used to set system operating parameters, process user input (UI) 306 on the GUI 304, and display gathered information such as images of ocular structures.” *Id.* at 4:5–9.

UF light beam 6 proceeds towards the patient’s eye 68, passing through various plates, mirrors, detectors, and control devices. *Id.* at 4:10–51. Beam combiner 34 transmits aim beam 202, OCT beam 114, and UF light beam 6. *Id.* at 4:51–54. UF light beam 6 proceeds through Z scan device 40 and then to x-y scan device 50. *Id.* at 4:59–5:9. Scanner 50 can automatically generate “aiming and treatment scan patterns . . . under the control of controller 300.” *Id.* at 5:35–37.

“An optional contact lens 66, which can be any suitable ophthalmic lens, can be used to help further focus the optical beam 6 into the patient’s eye 68 while helping to stabilize eye position.” *Id.* at 5:53–56.

Profilometer 415 is distal to X-Y scanner 50 “to allow for a continuous unobstructed view of the cornea of patient’s eye 68.” *Id.* at 11:31–33. Profilometer 415 “may be a placido system, triangulation system, laser displacement sensor, interferometer, or other such device, which measures the corneal topography,” and “may be used to prescribe an astigmatic keratotomy to correct the shape of a patient’s cornea to diminish its astigmatism.” *Id.* at 11:57–59.

Although contact lens 66 is shown in Figure 6, the ’024 patent explains that contact lens 66 or its disposition relative to cornea 406 of eye 68 may have to be modified, compensated for, or removed to suit the profilometer’s mode of operation. *Id.* at 11:36–48. “This is because profilometer 415 requires the cornea to be in its natural state, not forced into contact with a surface and possibly conforming to its shape, to accurately

measure cornea 406 and provide data to system 2 for calculation” *Id.* at 11:40–45.

F. Illustrative Claim

Petitioner challenges claims 1–17 and 20–26 of the ’024 patent.

Pet. 6. Of the challenged claims, claims 1, 8, 12, and 22 are independent.

Claim 1 is illustrative of the challenged claims and is reproduced below:

1. A cataract surgery method of treating target tissue in one or more of a cornea, limbus or sclera of a patient’s eye, comprising:
 - generating a treatment light beam;
 - deflecting the treatment light beam using a scanner to form first and second treatment patterns;
 - delivering the first treatment pattern to a first target tissue selected from the group consisting of the cornea, limbus and sclera of the patient’s eye to form a cataract incision that is sized to provide access to an eye chamber of the patient’s eye for lens removal instrumentation; and
 - delivering the second treatment pattern to a second target tissue to form a relaxation incision along or near limbus tissue or along corneal tissue anterior to the limbus tissue of the patient's eye to reduce astigmatism thereof,
- wherein the incision formed by delivering the first treatment pattern only partially extends through the target tissue.

Ex. 1007, 14:7–24.

G. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–17 and 20–26 are unpatentable on the following grounds (Pet. 6):

Claims Challenged	35 U.S.C. § ¹	Reference(s)/Basis
1–17, 20–26	103	Blumenkranz ² , Kurtz ³ , Weikert ⁴
Alternative: 16 ⁵	103	Blumenkranz, Kurtz, Weikert, Swinger ⁶
1–3, 6, 16, 17, 20–26	103	Kurtz, Swinger, Weikert
4, 5, 7–15	103	Kurtz, Swinger, Weikert, Benedikt ⁷

II. ANALYSIS

A. *Legal Standards*

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying

¹ The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. §§ 102 and 103, effective March 16, 2013. Because the latest possible effective filing date of the challenged claims of the ’024 patent is before this date (August 7, 2012), the pre-AIA version of these statutes apply. *See* 35 U.S.C. § 100(i)(2); Ex. 1007, code (22).

² US Patent Publication No. 2006/0195076 A1, published August 31, 2006. Ex. 1017 (“Blumenkranz”).

³ US Patent Publication No. 2008/0058777 A1. Ex. 1018 (“Kurtz”).

⁴ Mitchell P. Weikert and Douglas D. Koch, *Refractive Keratotomy: Does It Have a Future Role in Refractive Surgery?*, Cataract and Refractive Surgery (2005). Ex. 1019 (“Weikert”); *see* Ex. 1001 ¶ 73.

⁵ Petitioner challenges claim 16 as obvious over Blumenkranz, Kurtz, Weikert, and Swinger as an alternative ground, should the Board find that claim 16 would not have been obvious over the combination of Blumenkranz, Kurtz, and Weikert. Pet. 6, 45–46.

⁶ US 6,325,792 B1, issued December 4, 2001. Ex. 1021 (“Swinger”).

⁷ US Patent Publication No. US 2004/0066489 A1, published April 8, 2004. Ex. 1020 (“Benedikt”).

factual determinations, including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) if in the record, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. Level of Ordinary Skill in the Art

In determining the level of skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner contends one of ordinary skill in the art “would have had a Ph.D. in Physics, Biomedical Engineering, or a related science, such as Optical Engineering, or at least five years of experience in research, manufacturing, or designing medical optics or medical lasers.” Pet. 24–25. According to Petitioner, “[i]n either case, a [person of ordinary skill in the art] would have also had a moderate understanding of ophthalmology, and refractive and cataract surgery.” *Id.* at 25.

Patent Owner contends the field of invention is “ophthalmic surgical procedures and systems” and that the team of inventors that developed the ’024 patent included an ophthalmic surgeon. PO Resp. 9 (citing Ex. 1007, 1:15–16, 1:57–60; Ex. 2055). Thus, according to Patent Owner, the correct level of skill in the art includes meaningful experience with ophthalmic surgery, such as an ophthalmic surgeon with experience with medical optics or lasers, or an engineer with a Bachelor’s degree in a laser-related engineering or optics field who worked with an ophthalmic surgeon. *Id.* at

10 (citing Ex. 2062 ¶ 80). Patent Owner contends this definition of an ordinarily skilled artisan stands in contrast to the experience of Petitioner's declarant, Dr. Lubatschowski, who lacks experience in ophthalmic surgery. *Id.* at 10–11.

In its Reply, Petitioner argues that Patent Owner's definition of the ordinarily skilled artisan includes an engineer who “worked with a clinician,” and Dr. Lubatschowski has over 20 years of experience in laser applications for ophthalmology working with ophthalmic surgeons. Pet. Reply 4 (citing Ex. 2062 ¶ 80).

Both parties present compelling evidence that those of skill in the art would include individuals with experience in ophthalmology, as well as in the research, manufacture, and design of medical optics or medical lasers. Pet. 24–25; PO Resp. 10. The evidence of record also demonstrates that few individuals actually had such experience. Rather, teams of individuals with different expertise and various academic degrees would work together to design laser surgery systems intended to make incisions in the cornea, limbus, sclera, or lens of the eye.

Thus, we find that the person of ordinary skill in the art would have been an ophthalmic surgeon or someone with a Ph.D., M.S., or B.S. degree in physics, biomedical engineering, or a related science, such as optical engineering, and having experience in researching, manufacturing, or designing medical optics or medical lasers (e.g., 5 years of experience for a Ph.D. or more to compensate for lesser degrees). We further find that such persons would have worked in collaboration with one another to fill any necessary gaps in knowledge (e.g., the engineer would consult the medical doctor on clinical issues or physiology and the medical doctor would consult

the engineer on technical issues). This definition marries the two definitions proposed by the parties.

We note that although the level of ordinary skill in the art could include individuals with varying backgrounds and experience levels, where a declarant has knowledge of ophthalmic surgery but not medical optics, or of medical optics and engineering but not ophthalmic surgery, we will take this relative lack of direct experience or knowledge into account when weighing each declarant's testimony.

C. *Claim Construction*

In this proceeding, the claims of the '024 patent are construed "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b)." 37 C.F.R. § 42.100(b). Under that standard, the words of a claim are generally given their "ordinary and customary meaning," which is the meaning the term would have had to a person of ordinary skill at the time of the invention in the context of the entire patent including the specification. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Petitioner provides constructions for the terms "[a] cataract surgery method" and "the first and second treatment patterns are delivered simultaneously." Pet. 6–10.

Patent Owner provides constructions for the terms "[a] cataract surgery method" and "cataract incision . . . only partially extends through the target tissue." PO Resp. 12–14.

We address the construction of "[a] cataract surgery method" below. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)) ("[O]nly those terms need be

construed that are in controversy, and only to the extent necessary to resolve the controversy.”)).

“A cataract surgery method”

1. Whether the Preamble is Limiting

The preamble of independent claims 1, 8, 12, and 22 recites “[a] cataract surgery method of treating target tissue in one or more of a cornea, limbus, or sclera of a patient’s eye.” Ex. 1007, 14:7–9, 14:49–15:3, 15:15–37, 16:20–40. Petitioner contends this preamble phrase is a non-limiting statement of intended use. Pet. 6 (citing *Arctic Cat Inc. v. GEP Power Prod., Inc.*, 919 F.3d 1320, 1328 (Fed. Cir. 2019)). Petitioner acknowledges that the Examiner amended the preamble from “[a] method” to “a cataract surgery method” during prosecution, but contends this still represents an intended use that “fails to impart any patentable weight to the claimed method steps, which can be performed by multifunctional ophthalmic-surgery systems.” *Id.* at 6–7.

Patent Owner contends the preamble phrase is limiting because it was added by the Examiner to secure allowance of the claims. PO Resp. 12. In particular, Patent Owner notes that to overcome the prior art the Examiner suggested amending the claims “to exclude the other types of treatments,” replacing “[a] method” with “[a] cataract surgery method.” *Id.*

In its Reply, Petitioner asserts that “a closer look indicates the Examiner did not rely” on the insertion of the phrase “cataract surgery method” in the preamble when allowing the claims. Pet. Reply 5. Petitioner reasons that, although “the Examiner suggests the amendment was ‘to exclude other types of treatment,’ . . . the Notice of Allowance instead focuses on the fact that the prior art lacked *partially* penetrating cataract incisions.” *Id.* (citing Ex. 1009, 14). Thus, according to Petitioner, the

“Examiner’s (incorrect) conclusion that a partially penetrating cataract incision was patentable was the basis for allowance, not that the prior art failed to teach a ‘cataract surgery method.’” *Id.*

“[C]lear reliance on the preamble during prosecution to distinguish the claimed invention from the prior art transforms the preamble into a claim limitation because such reliance indicates use of the preamble to define, in part, the claimed invention.” *Catalina Mkt’g Int’l, Inc. v. Coolsavings.com, Inc.*, 289 F.3d 801, 808–09 (Fed. Cir. 2002). Here, and as discussed in further detail below, the Examiner expressly allowed the claims, in part, due to an Examiner’s Amendment changing “[a] method” to “[a] cataract surgery method.” Ex. 1009, 16. This reliance on the preamble to define the claimed invention transforms this portion of the preamble into a limitation. *See Catalina Mkt’g*, 289 F.3d. at 808.

2. *Claim Construction Analysis*

The determination that the preamble is limiting leads to the question of what a “cataract surgery method” entails. Patent Owner contends the term “cataract surgery method” “excludes methods for other treatments, such as Kurtz’s corneal transplant” procedures. PO Resp. 36. Petitioner disagrees, asserting that the term only requires a cataract or relaxation incision. Pet. Reply 6.

To construe a claim, we must look at the claim language, the written description, and if in evidence the prosecution history. *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996). This intrinsic evidence is “the most significant source of the legally operative meaning of disputed claim language,” and “in most situations an analysis of the intrinsic evidence alone will resolve any ambiguity in a disputed claim term.” *Id.* at 1582–83.

a) Claims

Independent claims 1, 8, 12, and 22 require “A cataract surgery method of treating target tissue in one or more of a cornea, limbus, or sclera of a patient’s eye.” Ex. 1007, 14:7–8, 14:49–3, 15:15–37, 16:20–40. Both parties agree that the term “cataract surgery” suggests that the system is intended for use in cataract surgery. Sur-Reply 3; Pet. Reply 6.

As noted above, cataract surgery involves incisions in the cornea, limbus, or sclera of a patient’s eye, as well as incisions in the lens capsule and lens. Pet. 18; Ex. 2063 ¶ 26. Although the term “cataract surgery” could imply any of these incisions, the preambles of claims 1, 8, 12, and 22 make clear that the intended purpose of the claimed cataract surgery method is to treat “one or more of a cornea, limbus, or sclera of a patient’s eye,” and the body of each independent claim is directed only to cataract and relaxing incisions.

Accordingly, the language of claims 1, 8, 12, and 22 suggests that the claimed “cataract surgery method” is used to treat tissue in the cornea, limbus, or sclera of a patient’s eye for purposes of cataract surgery, but does not suggest that the method must be used to also make unclaimed incisions in the lens capsule or lens (where the cataract would reside).

b) Written Description

The term “cataract surgery method” is not used in the written description of the ’024 patent. Review of the written description, however, provides useful insight into how the inventors viewed their disclosed invention. For example, in the Background of the Invention section, the written description focuses on the need for “ophthalmic methods, techniques, and apparatus to advance the standard of care of *corneal shaping* that may be associated with invasive cataract and other ophthalmic

pathologies.” Ex. 1007, 1:57–60 (emphasis added). The Summary of the Invention section then focuses on the use of patterned laser cutting in the cornea and limbus to form cataract and relaxation incisions. *Id.* at 1:64–66. In neither of these sections is laser surgery on the lens discussed or described. This suggests that the claimed method is directed to making incisions in the cornea, limbus, or sclera, and not necessarily the lens or lens capsule.

In at least one embodiment of the ’024 patent, it is noted that OCT data may be loaded into control electronics “and used to program and control the subsequent laser-assisted surgical procedure” on the lens and lens capsule. *Id.* at 6:14–16. This suggests that a *cataract surgery method* might be used to form incisions in the cornea as well as the lens. We are directed to no disclosure in the written description, however, indicating that the disclosed method must be used for all steps of cataract surgery, such as incisions in the lens or lens capsule.

Accordingly, we find that the written description supports that the claimed “cataract surgery method” is intended to provide incisions in the cornea, limbus, or sclera of a patient’s eye and could include, but is not required to include, incisions in the lens or lens capsule.

c) Prosecution History

In a Final Office Action during prosecution, the Examiner rejected the pending claims over the prior art combination of Blumenkranz (US Patent Publication No. 2006/0195076), Swinger (US Patent No. 6,325,792), and Bille (US Patent No. 4,907,586), as evidenced by Bille ’586 (US Patent No. 4,907,586). Ex. 1009, 58. In this rejection, the Examiner asserted that Blumenkranz discloses image guided laser surgery on the lens, as well as in other areas of the eye, such as sclera, iris, etc. *Id.* at 61. Given these

capabilities, the Examiner asserted that Blumenkranz could be used to provide multiple focus beams to create multiple incisions in the crystalline lens, as well as in the softer tissues of the other regions of the eye. *Id.*

The Examiner then noted that Swinger discloses applying multiple laser incisions in a patient's eye, as well as the utility of the method in corneal operations, such as keratoplasty, "correcting myopia, hyperopia, astigmatism, optical aberrations, etc." *Id.* at 61–62. The Examiner further noted that Swinger discloses making precise cuts in desired locations in a reproducible fashion, including "relaxation incisions," which can be made in any "predetermined length and depth." *Id.* at 62–63.

The Examiner then asserted that Bille teaches a laser beam guidance system that can focus on cells in the cornea with "great accuracy" and with control of the depth of the incision in the corneal portion of the eye. *Id.* at 64.

After summarizing the disclosures of Blumenkranz, Swinger, and Bille, the Examiner concluded that

it would have been obvious for one of ordinary skill in the art at the time of the instant invention was made to employ the system of Blumenkranz for not only making incisions in the lens but also for making incisions in the cornea by precisely controlling the location and depth of the incision such that the incision can penetrate epithelium, Bowman's membrane and only partially extend in to the stroma of cornea (col. 6, l. 40–68), [as] suggested by Bille.

Id. at 64.

In response to the Final Office Action, the Applicants amended the preamble of claim 1 to "identify the target tissues (to be treated)," i.e., "one or more of a cornea, limbus or sclera." *Id.* at 31 (claim amendment), 54 (applicant-initiated interview summary). Responding to the rejection over

Blumenkranz, Swinger, and Bille, the Applicants argued that although the prior art disclosed partially penetrating incisions “whose purpose is to **reshape the cornea**, it does not do so for *partially penetrating cataract incisions, whose purpose is not to re-shape the cornea.*” *Id.* at 41. The Applicants further argued that one of ordinary skill in the art would not have been motivated to generate a partial cataract incision because a partial incision is “unsuitable for its intended purpose of allowing access to the lens for surgical instruments.” *Id.*

In response to the Applicants’ amendments and arguments, the Examiner initiated an interview in which the Examiner suggested that “A method” in the preamble be amended to “A cataract surgery method” in each independent claim. *Id.* at 14. The Examiner stated as the reasons for allowance that the

Instant claimed method is distinguishable from those taught by the prior art of record in that the prior art does not teach or suggest the suitability of a delivery system that deliver[s] a first treatment pattern to a target tissue selected from the cornea, sclera, and limbus to form a cataract incision, which only partially extends through the target tissue, and provides access to the eye chamber of the patient’s eye without causing refractive changes. On the other hand, prior art of record teaches fully penetrating incisions for cataract surgery, which do not avoid refractive changes in accessing the crystalline lens during cataract surgery.

Id. at 17. Thus, the Examiner’s reason for allowance was not based on the ability of the claimed method to make incisions in both the lens and the cornea, but rather stems from the Examiner’s conclusion that the prior art of record did not disclose partially penetrating cataract incisions that do not cause refractive changes. *Id.*

In the subsequent Examiner-Initiated Interview Summary, the Examiner stated that the “Applicants presented the arguments that the prior art does not teach cataract incisions that partially penetrate the target tissue. Examiner suggested limiting the claimed method to cataract surgery to exclude the other types of methods.” *Id.* at 14. The claims then received a Notice of Allowance on September 24, 2015. *Id.* at 10.

Patent Owner argues that the Examiner’s explanation of amending the claims to “A cataract surgery method” in order to “exclude the other types of methods” demonstrates that the claimed method must be capable of making not only the claimed cataract and relaxing incisions, but also incisions in the lens. PO Resp. 12, 36.

The Examiner’s reasons for allowance regarding a “cataract surgery method” are ambiguous. For example, it is not clear how addition of the term “cataract surgery method” would exclude either Blumenkranz or Swinger, which both expressly disclose laser incisions in the lens or lens capsule for purposes of cataract surgery. Ex. 1017 ¶ 9; Ex. 1006, 76; Ex. 1021, 34:30–35:3. Bille does appear to be limited to laser incisions in the cornea, but the combination relied upon by the Examiner was for Blumenkranz to perform the claimed cataract and relaxation incisions. Thus, it is not clear how the Examiner’s Amendment to include a “cataract surgery method” would exclude any of the relied upon “surgery methods.”

In their response to the Final Office Action, the Applicants stressed that Blumenkranz, Swinger, and Bille do not individually or in combination teach or suggest a cataract incision, “which is intended to provide access to the crystalline lens for the lens removal

instruments during cataract surgery,” that only partially penetrates the cornea of the patient’s eye, and is not intended to “**reshape the cornea**.” Ex. 1009, 41. We are directed to no argument in this response that Blumenkranz, Swinger, or Bille must be capable of making incisions in both the lens and the cornea, or that the claimed incisions must be made during cataract surgery in which a laser is used to make incisions in both the cornea, limbus, or sclera, as well as the lens or lens capsule. Likewise, the Examiner’s stated reasons for allowance focused on the Applicants’ argument that the recited prior art references do not disclose a cataract incision in the cornea, limbus, or sclera of a patient’s eye that only partially extends through target tissue and does not cause refractive changes to the eye. *Id.* at 16–17.

In view of the foregoing, we understand that the addition of the term “cataract surgery method” was intended to make clear that the claimed method is used to make specific incisions related to cataract surgery, and not merely incisions “whose purpose is to **reshape the cornea**,” as disclosed for example in Bille. *Id.* at 64. We do not find that the prosecution history suggests that the term “cataract surgery method” requires that the method be used to form the claimed incisions in the cornea, limbus, or sclera, as well as unclaimed incisions in the lens or lens capsule.

d) Analysis

As discussed above, the intrinsic record of the ’024 patent, that is, the claims, written description, and prosecution history, suggests that a “cataract surgery method” is one that is used to create at least one incision related to cataract surgery. This may include cataract and relaxation incisions in the cornea, limbus, or sclera, or incision in the lens or lens capsule.

Accordingly, we construe the claim term “cataract surgery method” to require that the method be used to perform at least one incision related to cataract surgery, such as the expressly claimed cataract and relaxation incisions. Pet. 18–19 (Petitioner describing the steps of a “typical cataract surgery”).

D. Prior Art Status of Weikert

The Petition asserts that Weikert is an article, titled *Refractive Keratotomy: Does it Have a Future Role in Refractive Surgery?*, that was published in 2005 “as Chapter 14 in CATARACT AND REFRACTIVE SURGERY” and is therefore prior art to the ’024 patent under 35 U.S.C. § 102(b). Pet. 5, 26. In support of the Petition, Dr. Lubatschowski testifies that the identified chapter of Weikert was part of “the 2005 edition” of “CATARACT AND REFRACTIVE SURGERY.” Ex. 1001 ¶ 73.

In its authorized Preliminary Reply to the Preliminary Response, Petitioner provides a copy of the front cover of Weikert, as well as pages identifying the ISBN number, ISSN number, Library of Congress Control Number, and a 2005 copyright date for the reference. Preliminary Reply 1; Ex. 1060, 1–5.⁸ Petitioner also argues that a simple internet search of the citation provided in the Petition would provide the same information. Preliminary Reply 3.

Patent Owner contends that the Petition failed to provide evidence establishing Weikert is prior art to the ’024 patent and, therefore, “the Petition fails at the threshold.” PO Resp. 15. Patent Owner acknowledges that Petitioner submitted Supplemental Evidence and Reply arguments to

⁸ Here we reference the page numbers added in the bottom-right corner of the reference that were added by Petitioner.

support its assertion that Weikert is prior art to the '024 patent, but contends the Petition must establish a reference as prior art, and failure to do so in this case is fatal. *Id.*

A petition must “identify *with particularity* the grounds for institution and evidence supporting such grounds,” including “the prior art relied upon and evidence that it qualifies as such.” *Hulu, LLC v. Sound View Innovations, LLC*, IPR2018-01039, Paper 29 at 13 (PTAB Dec. 20, 2019) (precedential) (citing 35 U.S.C. § 312(a)). The Petition identifies the grounds for institution and the evidence supporting such grounds, and presents evidence that Weikert qualifies as prior art under 35 U.S.C. § 102(b). Pet. 5, 26. For example, Petitioner and Dr. Lubatschowski assert that CATARACT AND REFRACTIVE SURGERY “is a quarterly review series comprising chapters written by well-known specialists,” and that Weikert was included in the 2005 edition of CATARACT AND REFRACTIVE SURGERY as Chapter 14: *Refractive Keratotomy: Does it Have a Future Role in Refractive Surgery?* Ex. 1001 ¶ 73.

In addition, *Hulu* contemplates additional evidence being admitted in a reply to a patent owner preliminary response, as long as that evidence is responsive to the prior briefing. *Hulu*, Paper 29 at 14. In this case, Petitioner’s evidence submitted in its Reply is responsive to arguments made in prior briefs (i.e., Patent Owner’s Preliminary Response and Response), and simply confirms what was asserted in the Petition and Dr. Lubatschowski’s declaration, i.e., that Weikert is Chapter 14 of CATARACT AND REFRACTIVE SURGERY and the document bears a copyright date of 2005 (or, as asserted by Dr. Lubatschowski, is a “2005 edition”). Ex. 1060, 5, 12; Pet. 5, 26–27; Ex. 1001 ¶ 73. In addition, this evidence indicates that CATARACT AND REFRACTIVE SURGERY was published by

“Springer,” which is a well-known publishing company, and is the type of document that would be expected to be made publicly accessible. *See* Ex. 1001 ¶ 73 (asserting that CATARACT AND REFRACTIVE SURGERY “is a quarterly review series comprising chapters written by well-known specialists”); Ex. 1019, 220, 224, 227, 228, 230, 232 (providing a “Summary for the Clinician” at the end of several sub-chapters); Ex. 1060, 4–5.

The information presented in the Petition, as confirmed by the Preliminary Reply evidence submitted by Petitioner, sufficiently demonstrates that Weikert is prior art to the ’024 patent.

*E. Claims 1–3, 6, 16, 17, and 20–26 as
Obvious over Kurtz, Swinger, and Weikert*

Petitioner contends the subject matter of claims 1–3, 6, 16, 17, and 20–26 would have been obvious over the combined disclosures of Kurtz, Swinger, and Weikert. Pet. 47–57.

1. Kurtz

Kurtz discloses a system and method for resecting corneal tissue using a surgical laser. Ex. 1018, code (57). In particular, Kurtz discloses a system and techniques for transplanting corneas. *Id.* ¶ 2.

Kurtz explains that traditional techniques used for performing penetrating keratoplasty involved using a full-thickness cylindrical cut in both the recipient and donor corneas to resect corneal tissue. *Id.* ¶ 4. The resected donor tissue is then grafted into the recipient cornea in the same operating room and within minutes of the resection. *Id.*

Kurtz explains that femtosecond surgical lasers were previously used to create full thickness corneal incisions, but such systems have the drawback of taking up “valuable space within the operating room.” *Id.* ¶ 5. Given this drawback, Kurtz discloses that “[a]s an alternative, the

femtosecond surgical laser could be placed in a surgical preparation room.”

Id. In that scenario, extreme care must be taken not to expose the internal tissues of the cornea to contaminants “during the process of transferring the recipient and the donor tissue to the operating room for completion of the procedure.” *Id.*

To overcome these limitations, Kurtz discloses having the pulsed laser beam skip portions of the resection pattern, thereby leaving uncut gaps in the to-be-resected cornea. *Id.* ¶ 7. Kurtz explains that by leaving uncut gaps in the resection pattern, tissue along the incision and the internal chambers of the eye remain protected and unexposed to environmental contaminants, allowing the patient to be moved between the preparation room and the operating room without exposing the patient to contamination risks. *Id.* ¶ 14. Once in the operating room, the uncut gaps may be incised by the surgeon using an alternate surgical instrument, preferably a bladed instrument. *Id.* ¶¶ 8, 15.

Figure 2 of Kurtz is reproduced below:

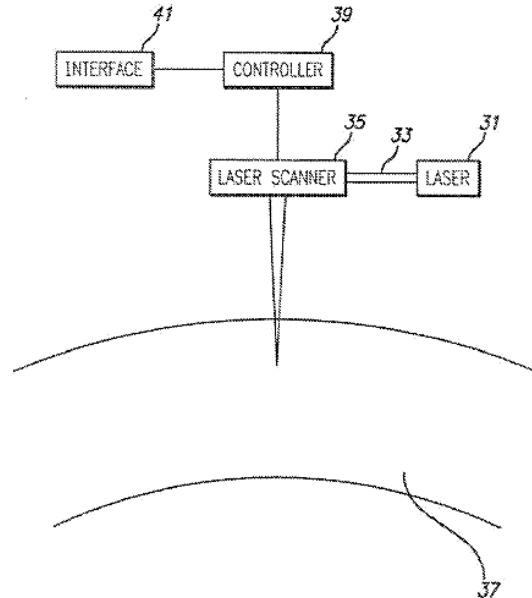


FIG. 2

Figure 2 of Kurtz is a schematic illustration of a system for resecting corneal tissue using a resection pattern having an uncut gap. *Id.* ¶ 12. In the embodiment shown in Figure 2, femtosecond surgical laser 31 generates a pulsed laser beam 33 and directs that beam into focusing assembly 35, “which in turn focuses the pulsed beam 33 into the cornea 37.” *Id.* ¶ 19. Interface 41 “presents the surgeon with several incision patterns from which the desired resection pattern is selected.” *Id.* Controller 39 “is a programmable computer which precisely controls the location of the beam focal point within the cornea 37 according to parameters received from the surgeon interface 41.” *Id.*

2. Weikert

Weikert reviews the history, use, and potential future of refractive keratotomy, which involves making incisions into the cornea of the eye,

often to correct astigmatism. Ex. 1019, 217.⁹ Weikert explains that the first clinical use of keratotomy to correct refractive error occurred in 1885, where a penetrating limbal incision was used to decrease astigmatism following cataract surgery. *Id.* (section 14.2). Although by the late 1990s laser-based systems “had replaced refractive keratotomy as the dominant technique for the surgical correction of refraction error,” Weikert notes that “incisional corneal surgery remains a useful tool in the surgeon’s repertoire of refractive procedures.” *Id.* at 218.

Weikert notes that clear corneal incisions (CCIs) “made during cataract surgery have been known to induce astigmatism by flattening the meridian on which the incision is centered.” *Id.* at 227 (section 14.7.1). “The amount of this surgically induced astigmatism (SIA) varies with incision length and placement.” *Id.* Weikert reports that one study comparing incision sizes of 3.2 mm, 4.0 mm, and 5.2 mm, found that the mean SIA was 0.09 D, 0.26 D, and 0.54 D, respectively. *Id.* In view of the various studies on the subject, Weikert reports that “0.0–0.5 D of SIA can be expected from temporal CCIs less than or equal to 3.2 mm.” *Id.* at 228.

Weikert notes that one method of correcting the astigmatism caused by corneal incisions for cataract surgery was to provide “a similar incision placed opposite to the temporal CCI,” with cataract surgery being performed only through one wound. *Id.* (section 14.7.2). Although such a procedure can reduce astigmatism, its “range is limited” and “carries [the] additional risk associated with the extra penetrating corneal wound.” *Id.* To correct higher levels of astigmatism, Weikert reports that “[p]artial thickness, arcuate or transverse corneal incisions” may be used and that “[a]rcuate

⁹ Our citations are to the original page numbers of the document.

incision have been combined with cataract surgery to reduce pre-existing astigmatism.” *Id.* at 228–229 (section 14.7.3).

In its conclusion, Weikert reports that “[a]s advances continue in the areas of intraocular lens design, crystalline lens removal and excimer laser refractive surgery, we are likely to see further decline in the use of refractive keratotomy.” *Id.* at 232.

3. Swinger

Swinger discloses the use of low energy, ultra-short (femtosecond) pulsed laser radiation to ablate ocular tissue in a controlled fashion. Ex. 1021, code (57). Swinger explains that the disclosed photodisruption process is gentle enough that it may be used for surgical procedures that were previously impossible using laser radiation, including “radial and arcuate keratotomy,” “capsulectomy, capsulorhexis, and phacoablation.” *Id.*

Figure 6 of Swinger is reproduced below:

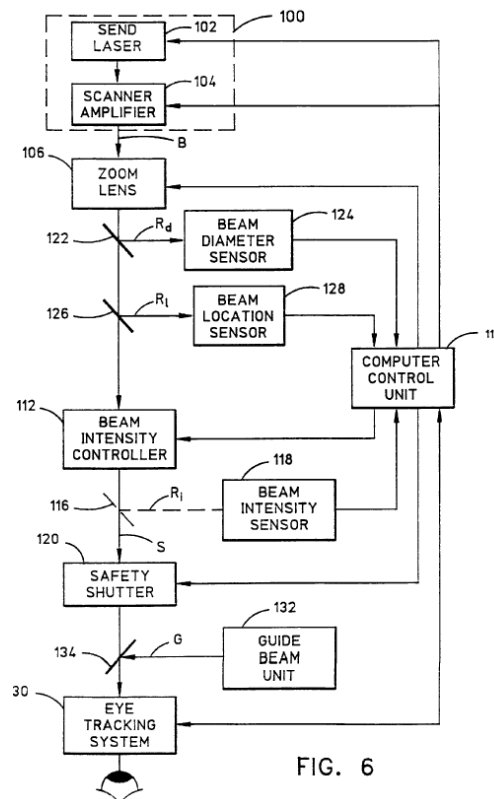


FIG. 6

Figure 6 is a block diagram of a preferred embodiment of the laser and control system of Swinger. *Id.* at 10:61–62, 17:1–30. As shown in Figure 6, laser unit 100 generates laser beam B. *Id.* at 17:1–2. Swinger explains that the preferred laser system includes a broad gain bandwidth laser using lasing ions such as titanium, chromium or neodymium and emitting at a preferred wavelength of 400 nm to 1900 nm, “which is generally transmissive in eye tissue.” *Id.* at 8:43–48.

Zoom lens 106 provides control over the diameter of laser beam B. *Id.* at 17:21–24. Beam-splitting mirrors 122 and 126 reflect part of the beam energy to beam diameter sensor 124 and beam location sensor 128, respectively. *Id.* at 18:43–45, 19:30–33. Beam intensity controller 112 is coupled to computer control unit 114, which is programmed to vary the intensity of surgical laser beam S, as necessary for a particular surgical procedure. *Id.* at 17:50–54. Safety shutter 120 is coupled to computer control unit 114 and is used to prevent unwanted or accidental laser radiation exposure of eye tissue. *Id.* at 18:10–24, 19:24–29. Guide beam unit 132 includes a low-power laser that provides a guide beam appropriate for direct viewing that is aligned with surgical laser beam S and acts as an indicator of the location of the treatment beam. *Id.* at 20:22–34.

Swinger discloses that its system “can easily create straight line and curved-line excisions, of any predetermined length and depth, at any location determined by a surgeon.” *Id.* at 20:49–51. One use of this system is “for performing radial keratotomies or making T-cuts or arcuate cuts, to correct myopia, hyperopia, or astigmatism (regular or irregular).” *Id.* at 21:21–23. Swinger explains that these cuts may be made using various laser scanning patterns and that these cuts may completely penetrate the cornea or may be made within the cornea. *Id.* at 33:7–17.

Swinger explains that capsulorhexis surgery may also be performed using the disclosed system as follows. *Id.* at 34:30–51. First, the focus of the laser beam spot is localized to the anterior lens capsule “by direct visualization using a visual HeNe laser beam focused to the same focal point as the ablating laser.” *Id.* at 34:52–55. “Then, the surgeon displaces the HeNe positioning beam just posteriorly to” the lens capsule and “photodisruption begins.” *Id.* at 34:58–61. According to Swinger, “[t]he cutting process can be totally computerized once the reference point on the capsule has been fixed, or the surgeon can terminate the process when the capsule has been visibly cut for 360 degrees.” *Id.* at 34:64–67.

4. *Analysis – Claim 1*

When combined, Petitioner contends Kurtz, Swinger, and Weikert teach or suggest every limitation of claim 1. In particular, Petitioner asserts that: (1) Kurtz discloses making openings in the cornea of a patient’s eye and Swinger teaches that systems similar to Kurtz’s can be used for steps of cataract surgery (limitation 1P) (*id.* at 48–49 (citing Ex. 1018 ¶¶ 7, 22; Ex. 1021, 8:55–67, 10:10–15)); (2) Kurtz and Swinger disclose generating a treatment light beam using a treatment light source (limitation 1.1) (*id.* at 49 (citing Ex. 1018 ¶¶ 19–20; Ex. 1021, 17:1–30, Fig. 6)); (3) Kurtz and Swinger disclose making incisions in the cornea of the eye that allow access to regions of the eye below the cornea, and that it would have been obvious to use partially penetrating incisions based on the disclosures of Kurtz “when the laser procedure is conducted in less-than-sterile environments” (limitations 1.3 and 1.5) (*id.* at 49–50 (citing Ex. 1018 ¶¶ 7, 8, 13–15, 19; Ex. 1021, 21:12–24, 33:7–22, Figs. 8B, 15W; Ex. 1001 ¶¶ 92, 315)); and (4) Weikert and Swinger disclose relaxation incisions to treat astigmatism (limitation 1.4) (*id.* at 50–51).

With respect to the reason to combine Kurtz, Swinger, and Weikert, Petitioner contends that Kurtz discloses a multifunctional ophthalmic-surgery system that can be used for corneal transplants, as well as “[o]ther applications,” and that Swinger discloses making incisions in the cornea, capsule, and lens in a predictable fashion that allows for safer methods in cataract surgery. Pet. 47 (citing Ex. 1018 ¶ 22; Ex. 1021, 10:10–16 (addressing the ability to provide a “smooth and regular opening” in eye tissue)). Although neither Kurtz nor Swinger expressly discloses partially penetrating cataract incisions, Petitioner contends that Weikert demonstrates that cataract and relaxation incisions have been known in the art for approximately 150 years, and that Kurtz does expressly explain the advantage of making partially penetrating incisions when the laser procedure is conducted in less-than-sterile environments. *Id.* at 48–50 (citing Ex. 1018 ¶¶ 7, 14, 19, Figs. 1A–H; Ex. 1001 ¶¶ 92, 315).

In view of these disclosures, Petitioner contends that one of ordinary skill in the art would have sought to use Kurtz’s system to deliver partially penetrating cataract and relaxation incisions in the cornea of a patient’s eye. *Id.* at 48.

Patent Owner contends Petitioner’s grounds based on Kurtz, Weikert, and Swinger fail because (1) Kurtz is a corneal surgery system and its corneal transplant methods would not be used to provide the type of cataract incisions used in cataract surgery, and (2) a person of ordinary skill in the art would not have modified Kurtz to perform surgery on the lens or lens capsule, which is required for a cataract surgery method. PO Resp. 36–43. We address both arguments below.

a) Providing Cataract Incisions Using the System of Kurtz

Patent Owner argues that the independent claims of the '024 patent were specifically amended during prosecution to exclude non-cataract surgery methods, such as Kurtz's corneal transplant methods. PO Resp. 36 (citing Ex. 1019, 14). And, "[g]iven that Kurtz is solely directed to corneal transplant surgery, and does not make any incisions in the lens," Patent Owner contends that "no skilled artisan would consider Kurtz to be a 'cataract surgery method.'" *Id.* at 37 (citing Ex. 2062 ¶¶ 248–249, 307; Ex. 2063 ¶¶ 52–57, 59–60).

Patent Owner further argues that Kurtz's "boilerplate statement" that "many more modifications are possible" fails to disclose a cataract surgery method, as defined by Patent Owner, "with sufficient precision and detail to establish that the subject matter existed in the prior art." *Id.* at 37–38 (quoting *Wasica Fin. GmbH v. Cont'l Auto. Sys.*, 853 F.3d 1272, 1284–1285 (Fed. Cir. 2017)).

Patent Owner's arguments are not persuasive because the term "cataract surgery method" does not require that the method be used to create incisions in the lens or lens capsule. Rather, a "cataract surgery method" is a method that is used to perform steps of cataract surgery, which can include cataract incisions and relaxation incisions. On this point, Petitioner persuasively explains why Kurtz is well-adapted to make cataract and relaxing incision in the cornea of a patient's eye and why one of ordinary skill in the art would have sought to do so based on the combined disclosures of Kurtz, Swinger, and Weikert. *See* Ex. 1001 ¶¶ 160–165.

b) Modification of Kurtz to Perform Cataract Surgery

Patent Owner contends that Kurtz would not be modified for surgery on the lens because it is "unsuited" for this purpose. PO Resp. 38.

According to Patent Owner, Kurtz uses precise focal spots with low pulse energy, which are achieved by using a delivery system with a high numeric aperture (NA), whereas lens surgery systems use a high pulse energy and larger focal spot to provide incisions in the lens. *Id.* at 38–39.

Again, Patent Owner’s arguments are not persuasive because we do not construe the term “cataract surgery method” to require that the method also be used to make incisions in the lens. All that is required is that the method is used to provide incisions in the cornea, limbus, or sclera that are part of cataract surgery. And, for the reasons discussed above, Petitioner persuasively explains why one of ordinary skill in the art would have sought to use the method of Kurtz, Swinger, and Weikert to form cataract and relaxation incisions as part of cataract surgery.

Patent Owner further argues that one of ordinary skill in the art would not have used Kurtz’s laser surgery system to make cataract and relaxation incisions if that system was not also able to perform the other steps of cataract surgery, such as incisions in the lens. PO Resp. 41–42. According to Patent Owner, because laser surgery systems are large and expensive “skilled artisans would not want to be burdened with two different laser systems for performing different steps of a cataract surgery procedure,” especially when manual blades were “an effective and low-cost option” for forming cataract and relaxation incisions. *Id.* at 43.

Petitioner argues in reply that surgeons often perform some parts of cataract surgery manually and some parts with a laser, depending on what will “get the best clinical outcome for the patient.” Pet. Reply 17–18 (quoting Ex. 1074, 52:19–53:2). Petitioner further argues that insurance coverage and costs are a factor in determining whether to use manual or laser incisions, which would be selected by individual surgeons based on

their budget and the systems at their disposal. *Id.* at 18. Finally, Petitioner notes that at least one of Patent Owner’s own laser systems is marketed as an “ideal choice” for laser cataract surgery, even though it cannot make incisions in the lens. *Id.* (citing Ex. 1066, 2).

As discussed above, Petitioner persuasively demonstrates that the system of Kurtz is capable of making cataract and relaxation incisions in a patient’s eye, and explains why one of ordinary skill in the art would have sought to use Kurtz’s system to make such incisions. In addition, we find persuasive Petitioner’s argument that surgeons in possession of Kurtz’s laser surgery system would have seen a benefit to using this system to perform accurate and precise incisions in the cornea, limbus, or sclera of a patient’s eye, even if other portions of the cataract surgery had to be performed manually or with a different laser surgery system. *See Orthopedic Equip. Co., v. United States*, 702 F.2d 1005, 1013 (Fed. Cir. 1983) (explaining that the mere fact that two disclosed systems would not be combined by businessmen for economic reasons is not “telling on the issue of nonobviousness”). Finally, contrary to Patent Owner’s general assertions about costs, size of equipment, and procedure time, Patent Owner markets a commercial system for creating cataract and relaxation incisions that does not perform incisions in the lens.¹⁰ Ex. 1066, 2–4. As such, Patent Owner’s

¹⁰ As noted by Patent Owner, Exhibit 1066 discloses a system that was not publicly available prior to the priority date of the ’024 patent. Nevertheless, this document casts doubt upon Patent Owner’s asserted concerns regarding apparatus size, costs, and procedure time; considerations that do not appear to have changed since the earliest priority date of the ’024 patent. In any event, our conclusion regarding the combination of Kurtz, Swinger, and Weikert would not change if we did not consider the disclosures of Exhibit 1066.

counter-arguments related to costs and business considerations of individual surgeons are not persuasive.

c) Conclusion

For the reasons set forth above, Petitioner persuasively demonstrates that every limitation of independent claim 1 is taught or suggested in Kurtz, Swinger, and Weikert. Petitioner also provides a reasoned explanation supported by factual underpinnings as to why one of ordinary skill in the art would have combined these references to arrive at the subject matter of claim 1 with a reasonable expectation of success. *See* Ex. 1001 ¶¶ 160–165; Pet. 47–48. Accordingly, Petitioner demonstrates by a preponderance of the evidence that independent claim 1 would have been obvious over Kurtz, Swinger, and Weikert.

5. Analysis – Claims 2, 3, 6, 17, 20–24, and 26

Petitioner also contends that Kurtz, Swinger, and Weikert disclose the subject matter of claims 2, 3, 6, 17, 20–24, and 26. Pet. 51–57. In support, Petitioner presents evidence that: Kurtz, Swinger, and Weikert teach or suggest delivering a second treatment pattern that is anterior to the limbus (i.e., in the cornea) (claim 2) (*id.* at 51 (citing Ex. 1001 ¶ 317)); Kurtz and Weikert teach or suggest providing first and second treatment patterns simultaneously (claims 3 and 24) (*id.* at 52 (citing Ex. 1018 ¶ 8; Ex. 1019, 13, 15–16; Ex. 1021, 34:61–64, 36:3–5)); Swinger discloses generating an aiming beam, deflecting this beam to form an aiming pattern, and delivering the aiming pattern to the target tissue to visually indicate a position of at least one of the treatment patterns (claim 6) (*id.* at 52–53 (citing Ex. 1021, 33:58–62, 34:27–28, 34:52–35:3, 35:50–57, 36:20–31, Fig. 15Y)); Weikert discloses using relaxation incisions to correct surgically-induced astigmatism (claim 17) (*id.* at 53–54 (citing Ex. 1019, 1, 2, 11, 12; Ex. 1001

¶¶ 118, 324)); Kurtz, Swinger, and Weikert teach or suggest forming a cataract incision at a limbal angle of about 30 degrees as seen from directly above the eye (claim 20) (*id.* at 42–43, 54 (citing Ex. 1021, 32:30–34, 32:60–63; Ex. 1018, Figs. 1A–1H; Ex. 1001 ¶ 325; Ex. 1052, 7; Ex. 1007, 10:66–11:1 (the ’024 patent disclosing that “[s]tandard cataract incisions typically require about 30 degrees of limbal angle as seen from directly above the eye”))); Weikert discloses that a cataract incision should preferably “not be formed at or near a steep axis of the cornea”) (claim 21) (citing Ex. 1019, 14–15; Ex. 1001 ¶ 301)); Kurtz, Swinger, and Weikert disclose each limitation of independent claim 22, including partially penetrating relaxation incisions (*id.* at 55–56 (citing Ex. 1021, 21:4–24, 33:7–22)); Kurtz and Swinger disclose treating target tissue in the cornea, limbus, or sclera of a patient’s eye (claim 23) (*id.* at 56); and Kurtz discloses the use of a bevel feature, which was a well-known, self-healing incision shape (claim 26) (*id.* at 57 (citing Ex. 1018 ¶¶ 16, 17, Figs. 1A–1F; Ex. 1001 ¶¶ 333–334)).

Other than the arguments addressed above concerning independent claim 1, Patent Owner does not address Petitioner’s arguments or evidence with respect to claims 2, 3, 6, 17, 20–24, and 26.

Upon review of Petitioner’s arguments and supporting evidence, we determine that Petitioner demonstrates by a preponderance of the evidence that claims 2, 3, 6, 17, 20–24, and 26 would have been obvious over Kurtz, Swinger, and Weikert.

6. Analysis – Claim 16

Claim 16 depends from claim 1 and further requires that the relaxation incision is formed from the inside of the target tissue towards the outside of

the target tissue, “without extending to the outside of the target tissue.”
Ex. 1007, 15:50–53.

Petitioner contends that Swinger discloses incisions that are formed from the inside of the target towards the outside of the target tissue, without extending outside the target tissue, and Kurtz discloses delivering partial incisions that start at the posterior surface but do not penetrate the anterior surface. Pet. 53 (citing Ex. 1021, 25:32–37, 30:60–63, 32:1–12, 33:46–51, 36:3–5; Ex. 1018, Figs. 1A, 1B, 1D, 1E, 1H). Petitioner asserts that it would have been obvious in view of either or both of Swinger’s and Kurtz’s disclosures “to apply relaxations to the posterior surface of the target tissue, as the same biomechanical laws apply, but it leaves the anterior surface of the cornea completely intact, reducing the risk of any accidental tears along the relaxation incision.” *Id.* (citing Ex. 1001 ¶ 323).

Patent Owner contends that the incision recited in claim 16 is an “intrastromal” relaxation incision that starts within corneal tissue and ends within corneal tissue. PO Resp. 51–52. In contrast to such incisions, Patent Owner contends Petitioner’s proposed incision begins at the posterior surface of the cornea, not “from the inside of target tissue.” *Id.* at 52. Patent Owner further contends that one of skill in the art would not have considered a posterior-facing relaxation incision to be interchangeable with prior art incisions, which extend to the anterior surface of the cornea, because the posterior surface of the cornea is lined with endothelial cells, which are prone to damage. *Id.* at 52–53.

In its Reply, Petitioner argues that the ’024 patent is silent on “intrastromal incisions,” and instead refers to incisions that are posterior incisions. Pet. Reply 24 n.10 (citing Ex. 1007, 13:15–20, 3:26–34). In any event, Petitioner contends that Swinger expressly describes “intrastromal”

incisions for correction of astigmatism, and contends that doctors Kang and Hatch agree that this incision type was known in the art. *Id.* at 24–25 (citing Ex. 1001 ¶ 76; Ex. 1021, 33:15–22, Figs. 15R, 15S, 15W, 15X; Ex. 1073, 136:23–140:13; Ex. 1074, 149:9–151:8).

Patent Owner responds that the '024 patent uses the term “inside” to refer to within the tissue and the term “outside” to refer to the tissue’s surface. Sur-Reply 26 (citing Ex. 1007, 3:29–32, 13:15–20). Thus, Patent Owner maintains its position that the incisions discussed in the Petition are not the “intrastromal” incisions recited in claim 16. *Id.* at 26–27. Patent Owner also argues that it is too late for Petitioner to change course and cite to new portions of Swinger to show that intrastromal incisions were known in the art. *Id.* at 27 (“[Petitioner] elected to map the challenged claim to Swinger at column 32, lines 1–12. It is too late to change course.”).

The language of claim 16 is somewhat ambiguous, requiring that the relaxation incision be formed from the inside of the target tissue towards the outside of the target tissue, without extending to the outside of the target tissue. Ex. 1007, 15:50–53. It could reasonably be interpreted as including an incision starting anywhere inside the corneal tissue that is below the surface of the cornea and extends towards the outside of the cornea (intrastromal incisions). PO Resp. 51. Conversely, it could include an incision that starts below the outside of the cornea, such as at the posterior surface, and extends towards the outside of the eye. Pet 53; Pet. Reply 24 n.10. We need not resolve the precise meaning of this claim phrase, however, because the prior art of record clearly indicates that intrastromal incisions (as Patent Owner understands claim 16 to require) were commonly used in the art and known to be used to correct astigmatism in laser surgical systems.

For example, Swinger, in the same general section identified in the Petition, discusses “making various incisions or excisions for correcting refractive errors such as myopia and astigmatism.” Ex. 1021, 33:7–9. Swinger notes that, “[i]n this discussion, only several commonly used patterns are described, and it is obvious that any kind of incision may be made, other than the examples given.” *Id.* at 33:9–12. Swinger then proceeds to describe using intrastromal incisions to correct astigmatism. *Id.* at 33:12–22. Given Swinger’s disclosures, we agree with Petitioner that it was understood in the art that intrastromal incisions could be used to correct for astigmatism and that it would have been obvious to use such “commonly used” incisions in the combination of Kurtz, Swinger, and Weikert. Pet. Reply 24–25.

With respect to Patent Owner’s argument that Petitioner may not now change course and argue that Swinger discloses intrastromal incisions, as noted by Patent Owner, Petitioner did not argue in the Petition that intrastromal incisions to correct astigmatism were disclosed in Swinger, in view of its interpretation of the language of claim 16. Pet. 53. The Petition did assert, however, that Swinger discloses incisions that are formed from inside the target tissue towards the outside, and cited to the same general portions of Swinger that it relies upon in its Reply. *See id.*; Pet. Reply 24–25 (both papers citing to incision patterns in Figures 15 of Swinger). Because the Reply directly addresses arguments made in Patent Owner’s Response, and is directed to addressing arguments related to potentially ambiguous claim language, Petitioner’s citation to clear disclosure in Swinger that intrastromal incisions were known and “commonly used” in the art to correct astigmatism was appropriate.

In view of the foregoing, Petitioner demonstrates by a preponderance of the evidence that claim 16 would have been obvious over Kurtz, Swinger, and Weikert.

7. *Analysis – Claim 25*

Claim 25 depends from claim 23 and further requires “wherein the incisions formed by delivering the second treatment pattern leaves at least 200 μm of tissue thickness.” Ex. 1007, 16:46–48.

Petitioner contends that Weikert “teaches that the corrective effect of relaxing incisions depends on the length, depth, and number” of relaxing incisions. Pet. 56 (citing Ex. 1019, 2). Thus, according to Petitioner, “the depth of the incision is nothing more than a result-effective variable subject to routine optimization,” and one of ordinary skill in the art would have found it obvious to leave at least 200 μm of tissue thickness when such a depth achieved the desired astigmatism correction. *Id.* at 56–57 (citing MPEP 2144.04; Ex. 1001 ¶ 332).

Patent Owner contends nothing in the prior art teaches leaving a minimum amount of 200 μm for any incision, let alone a relaxation incision, and Weikert teaches that relaxation incisions during cataract surgery remove 80% or more of the target tissue thickness, which would leave about 120 μm or less of tissue thickness. *Id.* (citing Ex. 1019, 13–14). With respect to Petitioner’s result-effective variable argument, Patent Owner contends this only applies to a variable that was recognized as result-effective, and no prior art recognizes a connection between remaining tissue thickness and ongoing structural integrity. Sur-Reply 57–58. According to Patent Owner, although “the relationship between relaxation incision depth and astigmatism correction (Ex. 1019, 2)” was known in the art, this would not

“lead a skilled artisan to optimize incision depth to leave the claimed minimum 200 μ m tissue thickness.” *Id.* at 58.

Petitioner argues in response that it was understood in the art that “more tissue thickness provides more structural integrity” and that “there was a known direct relationship between the uncut tissue and resulting astigmatism correction.” Pet. Reply 26 (citing Ex. 1073, 145:15–146:24; Ex. 1001 ¶ 307). And, “[s]ince increasing thickness provides more structural support while decreasing the corrective effect,” Petitioner contends “the thickness of the uncut tissue *is* a result-effective variable,” which Patent Owner “does not even attempt to argue . . . provides some special or unpredictable result.” *Id.* (citing Ex. 1073, 146:15–24; Ex. 1001 ¶ 307).

Patent Owner argues in its Sur-Reply that no prior art reference “discloses a nexus between remaining tissue thickness and incision stability or even suggests reducing incision depth to less than 80% of the corneal thickness.” Sur-Reply 28–29. Indeed, Patent Owner contends Dr. Hatch couldn’t think of a scenario where ophthalmologists would form such an incision. *Id.* at 29 (citing Ex. 1074, 148:4–22).

The evidence of record demonstrates that it was understood in the art that corneal incision depth can affect structural integrity, with deeper incisions leading to a potentially larger decrease in structural integrity. *See* Ex. 1073, 145:20–146:19; Ex. 2033, 2. It was also understood that deeper corneal relaxation incisions lead to a larger correction in astigmatism. Ex. 1001 ¶ 307; Ex. 1019, 218; Ex. 2062 ¶ 343; Ex. 2063 ¶ 43. Thus, corneal incision depth was understood to be a result-effective variable. *See In re Applied Materials*, 692 F.3d 1289, 1297 (Fed. Cir. 2012) (“A

recognition in the prior art that a property is affected by the variable is sufficient to find the variable result-effective.”).

The range of corneal incisions disclosed in the art submitted by the parties is from 80 to 90% of corneal thickness to 50% of corneal thickness. Ex. 2063 ¶ 43 (citing Ex. 2033, 2 (noting the use of incisions between 90% and 50% of corneal thickness, and explaining that incisions greater than 90% risk corneal perforation and incisions of less than 50% “have little effect on the corneal curvature and power”); Ex. 1001 ¶ 124 (Dr. Lubatschowski noting disclosures of relaxing incisions with a depth between 50% and 80% of corneal thickness), ¶ 307). Dr. Lubatschowski testifies that the average thickness of the cornea is approximately 500 μm . Ex. 1001 ¶ 307. Thus, a 50% cut in the cornea would leave 250 μm of tissue thickness, i.e., “at least 200 μm of tissue thickness,” as recited in claim 25. *Id.* Accordingly, the range of tissue thickness cited in claim 25 is disclosed in, or overlaps with, the prior art. *See Applied Materials*, 692 F.3d at 1295.

Given that corneal incision depth is a result-effective variable and that the general conditions of the claim are disclosed in the art, we agree with Petitioner that the claimed thickness range is prima facie obvious. *See E.I. DuPont de Nemours & Co. v. Synvina C.V.*, 904 F.3d 996, 1006 (Fed. Cir. 2018). A patent owner may rebut the presumption of obviousness for a result-effective variable by, *inter alia*, showing that the claimed range provides unexpected results. *Id.* Here, there is no evidence that leaving more than 200 μm of tissue leads to any unexpected result in either astigmatic correction or structural integrity of the cornea. As such, Petitioner demonstrates by a preponderance of the evidence that claim 25 would have been obvious over Kurtz, Swinger, and Weikert.

8. *Conclusion with Respect to the
Combination of Kurtz, Swinger, and Weikert*

For the reasons set forth above, Petitioner demonstrates by a preponderance of the evidence that claims 1–3, 6, 16, 17, and 20–26 would have been obvious over Kurtz, Swinger, and Weikert.

F. *Claims 4–5 and 7–15 over Kurtz, Swinger, Weikert, and Benedikt*

Petitioner contends the subject matter of claims 4–5 and 7–15 would have been obvious over the combined disclosures of Kurtz, Swinger, Weikert, and Benedikt. Pet. 57–65.

1. *Benedikt*

Benedikt discloses an apparatus for detecting the surface topography of a cornea of an eye. Ex. 1020, code (57). The apparatus of Benedikt includes a Placido Topometer and a CCD array. *Id.* ¶ 29. In use, light in a known pattern is projected on a cornea and the reflected light is captured as an image by the CCD array. *Id.* ¶ 31. This “allows measurement [of] the surface of the cornea 24 within a few milliseconds by recording usually more than 8,000 measuring points.” *Id.* ¶ 32. Because the recorded information “does not supply any information from deeper section of the eye,” however, Benedikt discloses combining the Topometer with either a wave front analyzer or optical coherence tomography. *Id.* ¶¶ 15, 32.

Figure 3 of Benedikt is reproduced below:

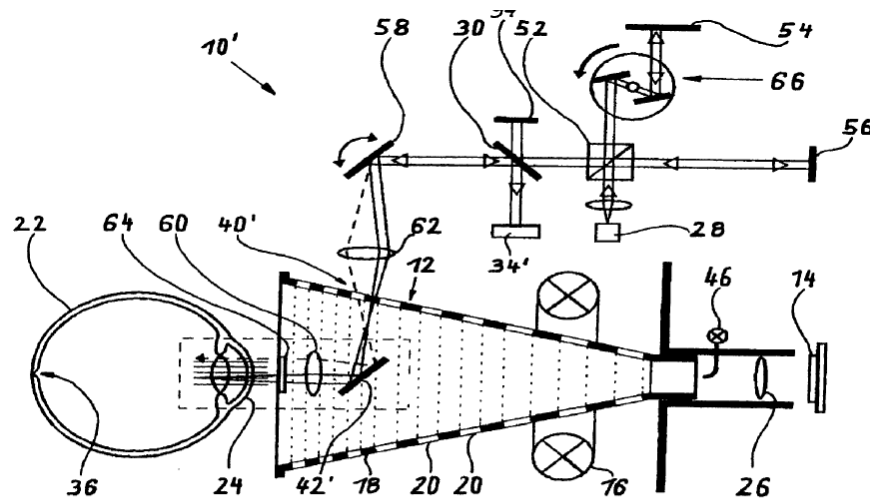


Figure 3

Figure 3 is a schematic representation “of an apparatus with a Placido Topometer and optical coherence tomography (OCT), with the optical coherence tomography being set up for acquiring tomographs from the anterior section of the eye.” *Id.* ¶ 25. In Figure 3, apparatus 10' consists of “a Placido Topometer with a Placido cone 12, a cone lamp 16 and a CCD array 14, as well as an optical coherence tomography.” *Id.* ¶ 41. Aperture 40' is provided on Placido cone 12, “through which the laser beam from the coherence tomography can be guided.” *Id.* The OCT device detects individual boundary surfaces of the eye, including the anterior and posterior surface of the cornea, anterior and posterior surface of the lens, and fundus of the eye. *Id.* ¶ 43.

In the disclosed design, “the measurements with the Placido Topometer and the coherence tomography can be performed either simultaneously or sequentially.” *Id.* ¶ 46. According to Benedikt, “the combination of Placido Topometry and coherence tomography leads to a

qualitatively novel and previously unachievable quantitative description of the eye in respect of diagnostics and therapeutics.” *Id.* ¶¶ 39, 46.

With respect to the use of a dual imaging system, Benedikt explains that a data record provided by the imaging system may be used “to introduce the individually optimal ablation pattern for the front surface of the cornea with photo-ablative lasers,” thereby automating the ablation of tissue and detaching “the ablation process from the surgeon’s manual dexterity.” *Id.* ¶ 39.

2. *Analysis – Claims 4, 5, 8–10, and 12–14*

a) *Claim 4*

Claim 4 depends from independent claim 1 and further recites a profilometer for measuring the surface profile of a patient’s cornea and a controller that “controls the formation of the second treatment pattern by the scanner in response to the measured surface profile.” Ex. 1007, 14:30–33.

Petitioner contends that Kurtz, Swinger, and Weikert “collectively teach a surgery scanning system for treating target tissue, including cataracts, in a patient’s eye,” but “do not expressly disclose a system with multiple detecting, imaging, and profiling subsystems.” Pet. 57. Petitioner asserts, however, that Benedikt recognizes that “an accurate understanding of the target anatomy is essential to ophthalmic surgery systems,” and provides this accurate illustration of the cornea and other structures of the eye by combining a topometer and OCT device. *Id.* at 58–59. Given the accurate and detailed information provided by the imaging system of Benedikt, Petitioner contends that one of ordinary skill in the art would have sought to integrate Benedikt’s imaging assembly into a laser treatment system, such as Kurtz and Swinger, “in order to plan and effect laser surgery with improved accuracy.” *Id.* at 59 (citing Ex. 1001 ¶¶ 169–170).

When Benedikt's topometer/OCT imaging device is used in combination with Kurtz, Swinger, and Weikert, Petitioner persuasively identifies where every limitation of claim 4 is taught or suggested in the prior art. *Id.* at 59–60.

Patent Owner contends Petitioner's combination arguments fail because (1) Benedikt's dual topometer/OCT imaging system is not more accurate than Swinger's direct visualization method; and (2) one of ordinary skill in the art would not have sought to use Benedikt's imaging system with prior art laser systems due to technical and design issues. PO Resp. 43–50. We address these arguments in turn.

(1) Accuracy of Benedikt's Imaging System

Patent Owner argues that Swinger uses a HeNe beam to accurately create treatment patterns and that Petitioner “offers no evidence that Benedikt's diagnostic imaging system would improve Swinger's accuracy or precision.” PO Resp. 46 (citing Ex. 2062 ¶¶ 269, 321). Petitioner further argues that the fact that Benedikt provides a “previously unattainable” illustration of the cornea would not have motivated a skilled artisan to use such a system, as the claimed “cataract surgery scanning system” “must be able to perform surgery on the *lens*.” *Id.* at 47 (asserting that Benedikt's imaging system is unable to provide topography of tissues below the cornea).

Benedikt describes its combined topometer/OCT imaging device as providing a “previously unattainable comprehensive topometrical/topographical illustration of the cornea.” Pet. 58; PO Resp. 46. As Kurtz uses its laser source to make incisions in the cornea, we find persuasive Petitioner's argument that one of ordinary skill in the art would have found it beneficial to implement Benedikt's imaging device to

aid the formation of cataract and relaxing incisions in the system of Kurtz, Swinger, and Weikert.

Patent Owner's counter-arguments are not persuasive because Swinger's method of direct visualization using a HeNe laser, although acceptable for surgery on the lens, does not provide the same level of topographical or topometrical data regarding the cornea as Benedikt, and does not provide a data record that can be used to automate laser surgery. Pet. Reply 22 (citing Ex. 1001 ¶¶ 29–31; Ex. 1020 ¶ 39). In addition, contrary to Patent Owner's arguments, we have found that the term "cataract surgery method" does not require that the method also be used to treat the lens.

(2) *Use of Benedikt's Imaging System in Kurtz and Swinger*

Patent Owner contends that Kurtz and Swinger both use an eye fixation device that is incompatible with Benedikt's diagnostic imaging system. PO Resp. 47. Patent Owner further contends that implementing a dual topometer/OCT device as part of a cataract surgery "has never been done" and "would have presented the difficult task of modifying optics to account for the different types of imaging beams." *Id.* at 49 (citing Ex. 2062 ¶¶ 231, 233, 325).

Petitioner argues in response that neither Kurtz nor Swinger requires eye fixation, and also that prior-art laser surgical systems with integrated imaging were known. Thus, according to Petitioner, a person of ordinary skill in the art would have known how to adjust the delivery system's optics and reprogram the controller, "including a calibration and registration between the treatment beam and imaging beams." Pet. Reply 23. In support of this position, Petitioner asserts that the '024 patent "offers no details" on how to program a controller, calibrate or register beams, or address

chromatic aberrations, strongly suggesting that these steps were within the skill of an ordinarily skilled artisan. *Id.* at 23–24.

In its Sur-Reply, Patent Owner argues that Kurtz’s and Swinger’s eye fixation devices are incompatible with a topometer and that OCT would provide information regarding “both the corneal surface profile” and subsurface structures, without the need for a topometer. Sur-Reply 23. And, given the “paltry benefits” of using Benedikt’s imaging system over Swinger’s HeNe aiming beam, Patent Owner contends that one of ordinary skill in the art would not have sought to overcome the considerable design challenges in aligning the imaging and treatment beams. *Id.* at 25.

For surgery on the cornea, whether used for diagnosis or therapy, Benedikt explains that a topographic scanner is particularly useful, as is a second scanner for detecting optical properties of layers of the eye disposed below the cornea. Ex. 1020 ¶¶ 3–8, 19, 32, 39. Consistent with this disclosure, and with Petitioner’s arguments in general, Dr. Hatch testifies that even when OCT data is available, she finds topometer data very useful. Ex. 1074, 246:6–248:16 (Dr. Hatch explaining that despite having OCT images, there is “a huge benefit to topography”). As such, we find persuasive Petitioner’s argument that one of ordinary skill in the art seeking to form a cataract incision and relaxing incisions in the cornea would have sought to implement a topometer, as disclosed in Benedikt.

As noted by Patent Owner, use of Benedikt’s topometer would require modification or removal of Kurtz’s eye fixation device. Petitioner persuasively demonstrates, however, that neither Kurtz nor Swinger require an eye fixation device. Pet. Reply 27 (citing Ex. 1018 ¶ 21; Ex. 1021, 23:35–56). Moreover, the ’024 patent merely states that “contact lens 66 or its disposition relative to cornea 406 of eye 68 may have to be modified, or

compensated for, to suit the profilometer's mode of operation," without providing any particular guidance on how to do so. Ex. 1004, 11:35–38. This suggests that the necessary modifications were not particularly difficult and were within the skill of a person of ordinary skill in the art.

Moreover, although integrating and co-registering Benedikt's imaging system with a treatment laser would present technical design challenges, the evidence of record suggests that such modifications were within the ordinary skill in the art. Pet. Reply 27; Ex. 1072, 76:21:77–3 (Dr. Kang testifying that in 2005 a person of ordinary skill in the art knew how to develop a system which could co-register the OCT and the laser beam). In any event, even if significant design challenges existed in implementing a dual topometer/OCT imaging system in Kurtz, given the advantages of Benedikt's imaging system for incisions in the cornea, we find that one of ordinary skill in the art would not have foregone or avoided the use of Benedikt's imaging system in light of such challenges.

For the reasons set forth above, we find that Petitioner persuasively demonstrates that Kurtz, Swinger, Weikert, and Benedikt teach or suggest every limitation of dependent claim 4. Petitioner also provides a sufficient explanation as to how and why one of ordinary skill in the art would have implemented Benedikt's dual imaging system in Kurtz, Swinger, and Weikert. Accordingly, Petitioner demonstrates by a preponderance of the evidence that claim 4 would have been obvious over Kurtz, Swinger, Weikert, and Benedikt.

b) Analysis – Claims 5, 8–10, and 12–14

Petitioner presents evidence and argument identifying where Kurtz, Swinger, Weikert, and Benedikt teach or suggest every limitation of claims 5, 8–10, and 12–14. Pet. 60–65. In particular, Petitioner presents evidence

that: (1) Benedikt discloses measuring scattering properties from different locations in a patient's eye and that the optical properties of the eye may be used to automate surgery on the cornea of a patient's eye (claim 5) (*id.* at 60–61 (citing Ex. 1020 ¶¶ 8, 10, 14, 16, 19, 39 42, 44, Figs. 3, 4)); Benedikt teaches or suggests delivering treatment patterns to at least one target tissue based upon a measured surface profile (independent claim 8) (*id.* at 62–63 (citing Ex. 1020 ¶ 39; Ex. 1001 ¶ 345)); Kurtz discloses providing first and second treatment patterns simultaneously and Weikert further discloses delivering cataract and relaxation incisions together (claims 9 and 13) (*id.* at Ex. 1018 ¶ 8; Ex. 1019, 13, 15, 16; Ex. 1001 ¶ 346)); Swinger teaches generating an aiming beam, deflecting this beam to form an aiming pattern, and delivering the aiming pattern to the target tissue to visually indicate a position of one of the first or second treatment patterns (claims 10 and 14) (*id.* at Ex. 1021, 33:58–62, 34:27–28, 34:52–35:3, 35:50–57, 36:20–31, Fig. 15Y)); and Kurtz, Swinger, Weikert, and Benedikt disclose every limitation of independent claim 12, including measuring scattering properties from different locations of a patient's eye and positioning a treatment pattern based on these measured properties (*id.* at 64–65 (citing analysis for dependent claim 5)).

Patent Owner does not directly address Petitioner's arguments with respect to claims 5, 8–10, and 12–14, beyond the arguments discussed above with respect to dependent claim 4.

Upon review of the parties' arguments and supporting evidence as a whole, we determine that Petitioner demonstrates by a preponderance of the evidence that claims 5, 8–10, and 12–14 would have been obvious over Kurtz, Swinger, Weikert, and Benedikt.

3. *Analysis – Claims 7, 11, and 15*

Claims 7, 11, and 15 depend from independent claims 1, 8, and 12, respectively, and further require capturing an image of target tissue, displaying the image of target tissue, and modifying a composition and location of at least one of the first and second treatment patterns in response to a user input device. Ex. 1007, 14:44–48, 15:11–15, 15:45–19.

Petitioner contends that Benedikt discloses capturing an image of target tissue, displaying the captured image, and modifying a composition and location of at least one of the treatment patterns in response to input received from a user input device. Pet. 61 (citing Ex. 1020 ¶¶ 4, 31, 32, Figs. 3, 4). Petitioner further contends that Kurtz discloses the use of a “programmable computer” with an interface to “present[] the surgeon with several incision patterns from which the desired resection pattern is selected.” *Id.* at 62 (citing Ex. 1018 ¶ 19). Given these disclosures, Petitioner contends one of ordinary skill in the art would have found it obvious to use Kurtz’s “programmable computer” to capture an image of target tissue, display the captured image, and allow the surgeon to modify the composition and location of at least one treatment pattern. *Id.* (citing Ex. 1001 ¶¶ 340–341).

Patent Owner contends Petitioner’s arguments with respect to claims 7, 11, and 15 are insufficient because Petitioner’s evidence “suggests modifying the shape of Kurtz’s treatment pattern, ***not its location.***” PO Resp. 62 (citing Ex. 2062 ¶¶ 291, 347) (Dr. Kang testifying that for all incision patterns presented in Kurtz, “the location of the incision pattern is the same, and the only thing that changes is the desired shape and gap placement”).

In response, Petitioner asserts that selection of a treatment pattern “necessarily selects and modifies the treatment pattern location,” and in any event Kurtz’s controller already “precisely controls the location of the beam focal point . . . according to parameters received from the surgeon interface 41.” Pet. Reply 28 (citing Ex. 1018 ¶¶ 19, 22, Figs. 1A–1H).

In its Sur-Reply, Patent Owner contends Petitioner is incorrect that selecting a treatment pattern necessarily selects and modifies the treatment pattern location, as “[a]ll of Kurtz’s incisions target a 360-degree incision in the central cornea.” Sur-Reply 28–29 (citing Ex. 2063 ¶ 60; Ex. 1073, 207:19–210:23).

Upon review of the parties’ arguments and supporting evidence, we find that Kurtz, Swinger, Weikert, and Benedikt teach or suggest every limitation of claims 7, 11, and 15, including capturing an image of target tissue, displaying that image to a user, and allowing a surgeon to modify at least one of the treatment patterns. We also credit the testimony of Dr. Lubatschowski that interface 41 of Kurtz would be used by a surgeon to alter both the shape and location of the claimed cataract and relaxation incisions, using the imaging provided by the system of Benedikt. Ex. 1001 ¶¶ 340–341.

We also agree with Petitioner that changing the shape of a particular treatment pattern would also change the location of the treatment pattern, and although the differences in shape and location may be minor between selected incision shapes when used for the corneal incisions expressly disclosed in Kurtz (as asserted by Dr. Kang), the claims do not require any particular amount of change in the location of the treatment pattern. Pet. Reply 28 (citing Ex. 1018 ¶ 22, Figs 1A–1H).

In view of the foregoing, Petitioner demonstrates by a preponderance of the evidence that claims 7, 11, and 15 would have been obvious over Kurtz, Swinger, Weikert, and Benedikt.

G. Claims 1–17 and 20–26 over Blumenkranz, Kurtz, and Weikert; and Claim 16 over Blumenkranz, Kurtz, Weikert, and Swinger

Petitioner contends that claims 1–17 and 20–26 would have been obvious over Blumenkranz, Kurtz, and Weikert, and that claim 16 would also have been obvious in view of Blumenkranz, Kurtz, Weikert, and Swinger. Pet. 28–47.

1. Blumenkranz

Blumenkranz is directed to a system and method for making incisions in eye tissue at different depths. Ex. 1017, code (57). The primary disclosed use of the system of Blumenkranz is for cataract surgery, with the disclosed system providing “rapid and precise openings in the lens capsule and fragmentation of the lens nucleus and cortex . . . using 3-dimensional patterned laser cutting.” *Id.* ¶¶ 3–11, 57, 69.

Figure 11 of Blumenkranz is reproduced below:

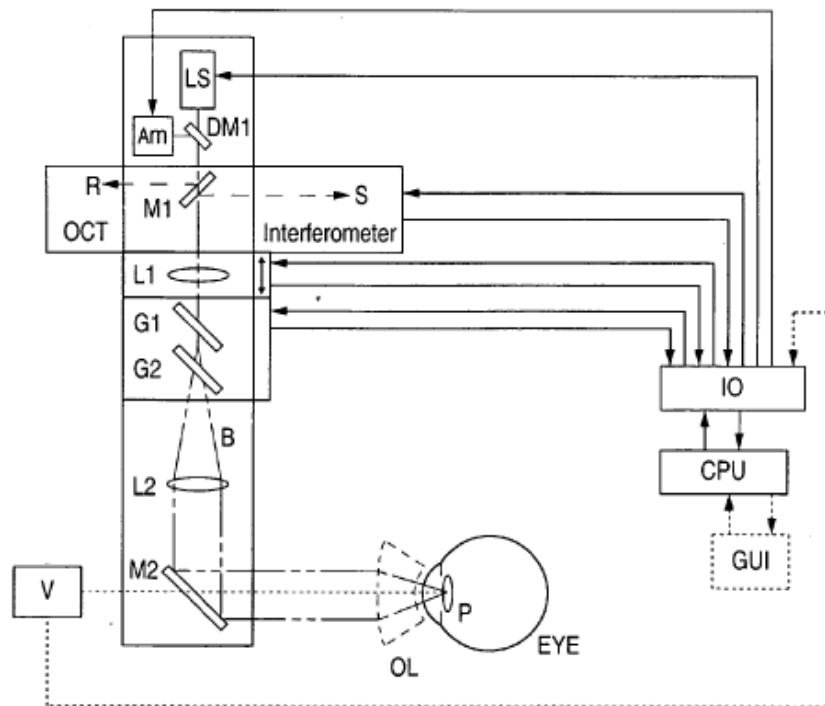


FIG. 11

Figure 11 is a plan diagram of one embodiment of Blumenkranz wherein the system projects or scans an optical beam into a patient's eye. *Id.* ¶ 34.

Figure 11 shows laser source LS and aiming beam source AIM having outputs that are combined using mirror DM1. *Id.* ¶ 75. In this configuration, laser source LS may be used for both therapeutics and diagnostics. *Id.* Mirror M1 serves to provide both reference input R and sample input S to an OCT Interferometer, which provides images to graphical user interface GUI. *Id.* ¶¶ 75, 77. For proper alignment of the treatment beam pattern, an alignment beam may be projected onto the target tissue with visible light. *Id.* ¶ 73. “This allows the surgeon to adjust the size, location, and shape of the treatment pattern.” *Id.* Cutting of ocular tissue is determined by scanning patterns that can be circular and spiral, with a vertical step similar to the length of the rupture zone. *Id.* ¶ 68.

Blumenkranz explains that although the primary discussion is of using the described system for capsulotomy and fragmenting the lens of the eye, the techniques described in the patent application “may be used to perform new ophthalmic procedures or improve existing procedures, including anterior and posterior capsulotomy, lens fragmentation and softening, dissection of tissue in the posterior pole (floaters, membranes, retina), as well as incisions in other areas of the eye such as, but not limited to, the sclera and iris.” *Id.* ¶ 71.

2. *Analysis – Claim 1*

Petitioner contends that Blumenkranz, Kurtz, and Weikert disclose every limitation of independent claim 1. In particular, Petitioner contends that Blumenkranz discloses a scanning system for cataract surgery that can “treat target tissue in one or more of the cornea, limbus, or sclera of a patient’s eye.” Pet. 30 (citing Ex. 1017 ¶¶ 8, 11, 21, 45, 71, 74). Petitioner further contends that the system of Blumenkranz has a treatment light source for generating a treatment light beam, as well as a scanner for deflecting the light beam to form treatment patterns under the control of a controller, with the scanner programmed to deliver multiple treatment patterns to target tissue, including the sclera of the patient’s eye, to form incisions. *Id.* at 30–31, 37 (citing Ex. 1017 ¶¶ 45, 50, 57, 73, Figs. 11–12). Petitioner concedes that Blumenkranz does not disclose delivering either a cataract incision that only partially penetrates the target tissue or a relaxation incision, but contends that such incisions are taught in Kurtz (partially penetrating incisions in the cornea) and Weikert (cataract and relaxation incisions). *Id.* at 31–34.

With respect to the reason to combine the various disclosures of Blumenkranz, Kurtz, and Weikert, Petitioner argues that one of ordinary

skill in the art would have sought to use partially penetrating cataract incisions in Blumenkranz in view of Kurtz's disclosure that such incisions protect the eye from environmental contaminants and infection when made in less-than-sterile environments. *Id.* at 28, 32–33 (citing Ex. 1018 ¶ 14). Petitioner further argues that one of ordinary skill in the art would have sought to use the system of Blumenkranz to form the relaxation incisions of Weikert because such incisions were well known in the art and using a laser to perform steps that were previously handled manually would have been obvious. *Id.* at 28–29 (citing *Leapfrog Enters., Inc. v. Fisher-Price, Inc.*, 485 F.3d 1157, 1161 (Fed. Cir. 2007) (“Accommodating a prior art mechanical device that accomplishes [a desired] goal to modern electronics would have been reasonably obvious to one of ordinary skill.”)).

Patent Owner contends Petitioner's arguments are unpersuasive because: (1) Blumenkranz does not teach or suggest the claimed incisions, and (2) one of ordinary skill in the art would not have modified Blumenkranz's laser system to make the cataract incisions disclosed in Kurtz and Weikert for multiple reasons. PO Resp. 20–34. We address Patent Owner's arguments below.

a) *Whether Blumenkranz Teaches or Suggests a Cataract Laser Designed to Cut the Claimed Tissue Types*

Patent Owner contends that, in its grounds based on Blumenkranz, Petitioner relies upon incisions in corneal tissue, yet Blumenkranz does not teach or suggest cutting corneal tissue, focusing instead on creating incisions “only in ocular structures well behind the cornea.” PO Resp. 20–21 (citing Pet. 31–32; Ex. 1001 ¶¶ 105–106, 108; Ex. 2062 ¶¶ 137–142). Patent Owner concedes that Blumenkranz mentions creating incisions in different tissue types, such as the sclera and iris, but contends that every structure

explicitly mentioned in the reference is located “well beneath the cornea,” which suggests that Blumenkranz does not contemplate creating incisions in corneal tissue. *Id.* at 21–24.

Petitioner argues in response that “the claims do not require corneal incisions, but [rather] incisions in ‘sclera’ tissue and ‘along or near limbus tissue.’” Pet. Reply 6–7 (citing Ex. 1007, 14:7–24 (claim 1)). And, because “Blumenkranz expressly teaches treating the *sclera* and *iris*,” Petitioner contends that “every claimed region is within the limits of Blumenkranz’s system.” *Id.* (citing Ex. 1017 ¶ 71; Ex. 2063 ¶¶ 14–15; Ex. 1074, 108:7–23, 104:4–15; Ex. 1070 ¶¶ 21–24).

To the extent the claims require incisions in the cornea, Petitioner contends one of ordinary skill in the art would have understood that, because Blumenkranz’s laser could treat the sclera and iris, it could also treat corneal tissue as well. Pet. Reply 7–8 (citing Ex. 1070 ¶¶ 21–24; Ex. 1069 ¶¶ 9–13). Petitioner reasons that the only corneal relaxation incisions discussed in the ’024 patent are at the periphery of the cornea (due to the desire to place these incisions close to the limbus/corneal/sclera junction) and, consequently, “the depths of the claimed corneal incisions are essentially identical to the depths of the sclera/iris.” *Id.* at 7–8. Thus, according to Petitioner, “Blumenkranz’s ability to cut sclera and iris tissue indicates it could also cut at the depths a [person of ordinary skill in the art] would have used for the claimed corneal incisions.” *Id.* at 9.

Patent Owner argues in its Sur-Reply that the Petition consistently asserted that the cataract incision and relaxation incisions would have been made in the cornea of a patient’s eye and, now in Reply, Petitioner has impermissibly shifted course to assert that the identified incisions could be made in limbal or scleral tissue. Sur-Reply 5–6 (citing Pet. 31–33). Patent

Owner further argues that relaxation incisions are typically formed in the “peripheral to mid peripheral cornea,” and by 2008 “sclera [cataract] incisions were not used at all.” *Id.* at 6 (citing Ex. 2065, 82:12–17, 223:14–225:11, 225:12–18; Ex. 1019, 14).

The evidence of record demonstrates that cataract incisions and relaxation incisions were well-known in the art. Ex. 1001 ¶¶ 23–24; Ex. 2063 ¶¶ 30, 33; Ex. 1019, 227; Ex. 1007, 12:24–26. For example, Weikert expressly discloses that corneal incisions are made during cataract surgery and that these incisions can “induce astigmatism by flattening the meridian on which the incision is centered.” Ex. 1019, 227. To treat this induced astigmatism, Weikert expressly discloses using relaxing incisions. *Id.* at 228.

The evidence of record also demonstrates that it was understood in the art that cataract incisions and relaxing incisions were known to be made in any of the cornea, limbus, or sclera of a patient’s eye. Ex. 1001 ¶ 22 (Dr. Lubatschowski testifying that cataract incisions may be made in the cornea, limbus, or sclera of a patient’s eye); Ex. 2004 ¶ 18 (Dr. Hatch testifying that “[d]uring cataract surgery, the surgeon makes a cataract incision in the overlying cornea, limbus, or sclera” of a patient’s eye); Ex. 2063 ¶ 33 (Dr. Hatch noting that cataract incision in the sclera were known, but are a “less common” form of incision); Ex. 1007, 12:24–26 (the ’024 patent expressly disclosing that “as is known in the art,” “relaxing incisions may be made in the limbus 408, or sclera 410”); *see Koninklijke Philips N.V. v. Google LLC*, 948 F.3d 1330, 1339 (Fed. Cir. 2020) (noting that it is appropriate to consider admissions in a patent’s specification “when assessing whether patent’s claims would have been obvious”).

Finally, we find that Petitioner sufficiently argued in its briefing that one of ordinary skill in the art would have used Blumenkranz's system to make cataract incisions in any of the cornea, limbus, or sclera of a patient's eye, as opposed to limiting its arguments to incision in the cornea. For example, the Petition expressly states that Blumenkranz discloses a delivery system that could be used to deliver a cataract incision in the sclera of a patient's eye. Pet. 31 (citing Ex. 1001 ¶¶ 68–71, 274). Dr. Lubatschowski also testifies in support of the Petition that a person of ordinary skill in the art “would have used Blumenkranz's system to deliver a first treatment pattern to the cornea, limbus, or sclera to form a cataract incision.” Ex. 1001 ¶ 274.

With respect to relaxing incisions, Dr. Lubatschowski testifies that the relaxation incision could be made “along or near limbus tissue” or in “other target tissue,” as opposed to “along corneal tissue.” Ex. 1001 ¶ 275; Pet. 33. The sclera is a tissue type that is not corneal tissue and is “near limbus tissue.” *See, e.g.*, Ex. 2062 ¶ 21 (Dr. Kang describing the tissues of the eye and their relative locations); Ex. 2063 ¶ 14 (Dr. Hatch describing the basic anatomy of the eye). Moreover, the '024 patent concedes that it was known in the art that relaxing incisions could be made in any of the cornea, limbus, or sclera of a patient's eye. Ex. 1007, 12:24–26.

As noted by Patent Owner, after discussing the various disclosures of Blumenkranz and Swinger, the Petition asserts that “[b]ased on these teachings, it would have been obvious [for a person of ordinary skill in the art] to use Blumenkranz's system to create a corneal incision that provides access for lens removal instrumentation.” Pet. 33 (citing Ex. 1001 ¶ 274); Sur-Reply 5. According to Patent Owner, this assertion led them to address only the question of whether Blumenkranz could be used for corneal

incisions, and not whether the system would have been used to form cataract incisions in the limbus or sclera. Tr. 113:20–115:6.

We agree that the Petition could have been clearer on this point. However, at the beginning of the section cited by Patent Owner, Petitioner expressly asserts that Blumenkranz could be used to target the sclera of a patient's eye to form a cataract incision. Pet. 31. Consistent with this disclosure, and in the section identified by Patent Owner, Petitioner cites repeatedly to Dr. Lubatschowski's testimony that a person of ordinary skill in the art would have used Blumenkranz's system to deliver a cataract incision in any of the cornea, limbus, or sclera. Ex. 1001 ¶ 274. These arguments in the Petition reasonably put Patent Owner on notice that Petitioner was relying on the ability of Blumenkranz's system to perform cataract incisions in any of the cornea, limbus, or sclera of a patient's eye. See Ex. 1001 ¶ 107 (Dr. Lubatschowski testifying that "another well-known aspect of cataract surgery involved incision in the cornea, sclera, or limbus to provide a surgeon access to the inner eye chamber").

As the sclera is a claimed tissue type for both types of incisions, and it is undisputed that Blumenkranz is capable of forming incisions in the sclera, we focus on this tissue type for the Blumenkranz-based grounds set forth in the Petition.

*b) Whether a Person of Ordinary Skill in the Art
Would have Modified Blumenkranz's Laser System*

Patent Owner argues that one of ordinary skill in the art would not have modified Blumenkranz to perform incision in the cornea because Blumenkranz's laser would have been unacceptable for such incisions. Patent Owner reasons that "Blumenkranz uses low focusing precision and

high laser energy, while corneal incisions require high focusing precision and low laser energy.” PO Resp. 25 (citing Ex. 2062 ¶¶ 90–95, 149–183).

This argument is not persuasive in this case because Petitioner is not relying only on incisions in the cornea. Rather, for the reasons set forth above, Petitioner contends cataract incisions and relaxation incisions could have and would have been made in any of the claimed tissue types, i.e., the cornea, limbus, and sclera.

c) *Forming Weikert’s Manual Incisions with Blumenkranz’s Laser*

Patent Owner contends that Petitioner’s proffered motivation for forming Weikert’s manual cataract incisions using Blumenkranz’s laser is that laser systems are “more accurate and precise.” PO Resp. 33 (citing Pet. 28–29). Patent Owner asserts that this argument fell apart during cross-examination, with Dr. Lubatschowski unable to identify “the advantages and disadvantages, from a clinical perspective, of using a blade versus a laser for a surface cut,” testifying that the benefits of using lasers compared to manual blade methods “depends on the situation,” and as a “non-ophthalmologist,” he did not want to “make any judgement calls.” *Id.* at 32–33 (quoting Ex. 2041, 107:8–22, 108:3–7, 109:9–15).

Patent Owner also contends Weikert’s manual incision methods provide “predicable and reproducible incision profiles,” and are “effective,” “low-cost, and low-risk.” *Id.* at 33. And, given the lack of evidence that laser incisions would be more accurate or more precise than manual methods, Patent Owner contends “[a] skilled artisan would have had no motivation to undertake the expensive task of redesigning Blumenkranz’s laser delivery optics.” *Id.* at 34.

Petitioner argues in its Reply that there was a consistent, acknowledged progression in the art from manual to more precise laser

methods, and there was “general excitement and optimism over how manual procedures could be improved by replacing them with lasers.” Pet. Reply 15. And, because lasers were known to perform some aspects of cataract surgery, Petitioner contends a person of ordinary skill in the art “would have been motivated to improve all aspects of cataract surgery, including Weikert’s relaxing incisions.” *Id.* at 13–14.

In its Sur-Reply, Patent Owner argues that the idea that skilled artisans would have used lasers for “all aspects” of cataract surgery is “[n]onsense,” as “[e]ven today, ophthalmologists perform the claimed incisions manually, even while performing other aspects of cataract surgery with a laser.” Sur-Reply 17–18 (citing Ex. 2065, 85:3–8; Ex. 1074, 50:17–55:4). And, given the “effective,” “low-cost,” “reproducible,” and “low-risk” manual blade methods discussed in Weikert, Patent Owner contends there was simply no need or desire to implement laser based surgical systems for these incisions. *Id.* at 18.

On this record, we find that there was a consistent progression in the art towards performing all aspects of eye surgery using lasers. For example, Weikert discloses that “the advent of excimer laser surgery has led to a tremendous decrease” in manual corneal incisions to correct refractive error. Ex. 1019, 217. Indeed, “[a]s technological advances continue” in various laser surgery techniques, including “crystalline lens removal,” Weikert asks: “Is there a future role for refractive keratotomy in modern refractive surgery?” *Id.*

We further find that the consistent push to perform various incisions in the eye using a laser was based on the view that such incisions are more precise and accurate than manual incisions performed by a surgeon. Consistent with this conclusion, Blumenkranz discloses a laser surgery

system that allows “rapid and precise openings in the lens capsule and fragmentation of the lens nucleus and cortex” using “3-dimensional patterned laser cutting.” Ex. 1017 ¶ 9. Similarly, Swinger discloses a femtosecond laser system that can perform corneal and lens procedures, and states that prior art mechanical and manual incisions in the cornea were unable to “precisely and reproducibly cut living corneal tissue with a minimum of trauma to the cornea.” Ex. 1021, 6:59–64, 8:55–57, 34:30–67, Fig. 15A1 (comparing the smooth contoured incision of the invention to the irregular opening created using manual techniques). In contrast to those manual techniques, Swinger discloses that its laser surgery system allows for “accurate control of tissue removal.” *Id.* at 9:1–6.

Consistent with the disclosures of the prior art, Dr. Lubatschowski testifies that laser-surgical incisions are “accurate and precise.” Ex. 1001 ¶¶ 91, 130, 177. Likewise, Dr. Hatch and Dr. Kang testify that femtosecond laser surgery allows for incisions to be performed in an accurate and “precise way.” Ex. 2063 ¶¶ 27, 60 (noting that Kurtz’s laser creates “a precise incision in the cornea”); Ex. 2062 ¶ 33 (Dr. Kang testifying that nanosecond laser pulses can provide “accurate, nonthermal ablation”). Thus, we find persuasive Petitioner’s argument that one of ordinary skill in the art would have sought to use Blumenkranz’s laser system to create cataract and relaxation incisions with improved accuracy and precision.

Often, cataract incisions and relaxation incisions are made in the cornea of the eye. Dr. Lubatschowski testifies, however, that it was understood in the art that access to the anterior chamber can be accomplished through incisions in any of the cornea, limbus, or sclera of the eye. Ex. 1001 ¶¶ 22, 117, 200. Likewise, Dr. Hatch testifies that cataract incisions in the sclera were known in the art. Ex. 2063 ¶¶ 30, 33 (noting that

cataract incisions in the sclera were known, but “less common”). As such, we find that one of ordinary skill in the art would have understood that Blumenkranz’s system was capable of making the corneal and relaxation incisions generally discussed in Weikert in the sclera of a patient’s eye, and would have sought to do so as this is a known tissue type in which to place cataract and relaxing incisions. *See* Ex. 1007, 12:24–26 (“Likewise, as is known in the art, such relaxing incisions may be made in the limbus 408, or sclera 410.”).

d) Conclusion—Independent Claim 1

In view of the foregoing, and upon review of the parties’ arguments and evidence as a whole, we find that Blumenkranz, Weikert, and Kurtz, when considered in light of the knowledge of one of ordinary skill in the art, teach or suggest every limitation of claim 1 of the ’024 patent. Petitioner also explains sufficiently why one of ordinary skill in the art would have sought to combine the disclosures of these references to arrive at the subject matter of claim 1 with a reasonable expectation of success. Thus, Petitioner demonstrates by a preponderance of the evidence that the subject matter of claim 1 would have been obvious over Blumenkranz, Kurtz, and Weikert.

3. Analysis – Claims 3–6, 8–10, 12–14, 17, 20–24, and 26

Petitioner identifies where Blumenkranz, Weikert, and Kurtz teach or suggest the subject matter of claims 3–6, 8–10, 12–14, 17, 20–24, and 26. In particular, Petitioner demonstrates that: Blumenkranz discloses delivering multiple treatment patterns simultaneously (claims 3, 9, 13, and 24) (*id.* at 34 (citing Ex. 1017 ¶ 101; Ex. 1001 ¶ 279)); Blumenkranz discloses measuring the surface profile of a surface of the cornea of the patient’s eye, and that a person of ordinary skill in the art would have used this data to control the formation of the second treatment pattern that is intended to treat

astigmatism (claim 4) (*id.* at 35–36 (citing Ex. 1017 ¶¶ 56, 57, 59, 68, 73–74, 78, 85; Ex. 1001 ¶¶ 282–283)); Blumenkranz discloses measuring scattering properties from different locations of a patient’s eye and using this information to position one of the cataract incision or relaxing incisions in a patient’s eye (claim 5) (*id.* at 36–37 (citing Ex. 1007, 6:58–67, 10:45–52; Ex. 1001 ¶ 284)); Blumenkranz discloses generating an aiming beam, deflecting the aiming beam to form an aiming pattern, and delivering the aiming pattern to target tissue to visually indicate a position of the cataract or relaxing incisions (claims 6, 10, and 14) (*id.* at 37 (citing Ex. 1017 ¶¶ 56, 75, 77, 84–85)); Weikert discloses the use of relaxation incisions to correct pre-existing astigmatism and/or surgically-induced astigmatism (claim 17) (*id.* at 41 (citing Ex. 1019, 1–2, 11–12; Ex. 1001 ¶¶ 118, 299)); Blumenkranz, Kurtz, and Weikert teach or suggest a first treatment pattern that is formed at a limbal angle of about 30 degrees as seen from above the eye (claim 20) (*id.* at 42 (citing Ex. 1017 ¶¶ 61–62, 68–69; Ex. 1018, Figs. 1A–1H; Ex. 1001 ¶ 300; Ex. 1007, 10:67–11:1 (the ’024 patent indicating that 30 degree angles are “standard” in cataract surgery))); Blumenkranz, Kurtz, and Weikert teach or suggest not forming the cataract incision at or near a steep axis of the cornea (claim 21) (*id.* at 42–43) (citing Ex. 1019, 14–15; Ex. 1001 ¶ 301)); Blumenkranz, Kurtz, and Weikert teach or suggest a cataract incision that is delivered to the cornea, limbus, or sclera of a patient’s eye (claim 23) (*id.* at 31–34, 44); and Kurtz discloses incisions that include a bevel feature, “which was also a well-known, self-healing incision shape” (claim 26) (*id.* at 45 (Ex. 1018 ¶¶ 16–17, Figs. 1A–1F; Ex. 1017 ¶ 99; Ex. 1001 ¶¶ 308–309)).

With respect to independent claims 8, 12, and 22, Petitioner identifies where Blumenkranz, Kurtz, and Weikert teach or suggest every limitation of these claims. Pet. 38–40, 43–44.

Patent Owner does not directly contest Petitioner’s arguments with respect to claims 3–6, 8–10, 12–14, 17, 20–24, and 26, beyond those discussed above with respect to independent claim 1.

Upon review of the arguments and evidence of record, we determine that Petitioner demonstrates by a preponderance of the evidence that claims 3–6, 8–10, 12–14, 17, 20–24, and 26 would have been obvious over Blumenkranz, Kurtz, and Weikert.

4. Analysis – Claims 7, 11, and 15

Claims 7, 11, and 15, depend from independent claims 1, 8, and 12, respectively, and further require “capturing an image of the target tissue, displaying the captured image, and modifying a composition and location of at least one of the first and second treatment patterns on the patient’s eye in response to input received from a user input device.” Ex. 1007, 14:44–48, 15:11–15, 15:45–49.

Pointing to the disclosures of paragraph 77, Petitioner argues that Blumenkranz’s system has a camera for capturing an image of the target tissue, a display device, and a graphic user interface for modifying the composition and location of a treatment pattern(s). Pet. 37–38 (citing Ex. 1017 ¶¶ 77, 88, Figs. 11, 12, 14).

Patent Owner argues in response that the GUI disclosed in paragraph 77 of Blumenkranz is a system for generically processing user input, and not a GUI that allows a user to modify a composition and location of a treatment pattern. PO Resp. 60–61. Patent Owner further argues that Figures 11, 12, and 14 of Blumenkranz merely depict the GUI generically,

with no disclosure of the capabilities of the GUI. *Id.* at 60. Patent Owner acknowledges that Dr. Lubatschowski also cites to Kurtz and Weikert for this claim limitation, but contends theories set forth in an expert declaration that are not incorporated or made in the Petition must be disregarded. *Id.* at 61.

Petitioner argues in Reply that there is no dispute that Blumenkranz discloses a GUI with the claimed functionality, as is expressly set forth in paragraph 77 of Blumenkranz, and that paragraph 73 of Blumenkranz confirms this by unambiguously disclosing that the GUI of Blumenkranz allows a surgeon to modify the composition and location of a treatment pattern. Pet. Reply 27–28 (citing Ex. 1073, 198:13–199:14; Ex. 1017 ¶¶ 73, 77, 88, Fig. 15; Ex. 1001 ¶ 288). And, although paragraph 73 of Blumenkranz is cited only in Dr. Lubatschowski’s declaration, Petitioner contends it is acceptable to cite additional paragraphs of a reference in a Reply to respond to arguments raised in the Patent Owner Response. *Id.* at 27 n.11.

In its Sur-Reply, Patent Owner asserts that this new paragraph of Blumenkranz (paragraph 73) may not be relied upon in the Reply but, in any event, it does not cure the deficiencies in the Petition because it only discusses an alignment beam/pattern, which Patent Owner contends is not a GUI. Sur-Reply 29 (citing Ex. 1073, 291:20–25, 280:4–281:3).

Blumenkranz discloses a camera, a display device, and a GUI. Ex. 1017 ¶ 77, Fig. 11, 12, 14. In Blumenkranz, an aiming beam may be directed to the patient’s eye and displayed on a screen, “such as a graphical user interface.” *Id.* ¶ 77. Blumenkranz also discloses that the alignment beam may allow the “surgeon to adjust the size, location and shape of the treatment pattern.” *Id.* ¶ 73. When considered in combination, we agree

with Petitioner that paragraphs 73 and 77 would indicate to one of ordinary skill in the art that the GUI of Blumenkranz allows a surgeon to “adjust the size, location and shape of the treatment patterns.” *Id.* ¶¶ 73, 77. Thus, we find that Blumenkranz discloses the subject matter of claims 7, 11, and 15.

We further find that Petitioner did not shift its argument in Reply and that it was not improper to point out, in responding to Patent Owner’s arguments, further disclosures of Blumenkranz—discussing the same general embodiment identified in the Petition—that confirm the assertion in the Petition that Blumenkranz teaches or suggests a camera, a display device, and a GUI that allows a surgeon to adjust the size and location of a treatment pattern. *See Apple Inc. v. Andrea Elects. Corp.*, 949 F.3d 697, 706–707 (Fed. Cir. 2020) (explaining that a petitioner may introduce new evidence in reply, as long as petition does not raise a new legal argument and is a legitimate reply to evidence introduced by the patent owner).

Accordingly, Petitioner demonstrates by a preponderance of the evidence that Blumenkranz, Kurtz, and Weikert teach or suggest every limitation of claims 7, 11, and 15.

5. *Analysis – Claim 25*

Claim 25 depends from claim 23, which depends from independent claim 22, and further requires “wherein the incisions formed by delivering the second treatment pattern leaves at least 200 μm of tissue thickness. Ex. 1007, 16:46–48.

Petitioner contends that Weikert “teaches that the corrective effect of relaxing incisions depends on the length, depth, and number” of relaxing incisions. Pet. 45 (citing Ex. 1019, 218). Thus, Petitioner contends “the depth of the incision is nothing more than a result-effective variable subject to routine optimization.” *Id.* (citing MPEP 2144.04). As such, Petitioner

contends it would have been obvious to one of ordinary skill in the art “to deliver a relaxation incision that left at least 200 μm of tissue thickness when such a depth achieved the desired astigmatism correction.” *Id.* (citing Ex. 1001 ¶ 307).

In response, Patent Owner relies on the same arguments discussed above with respect to the combination of Kurtz, Swinger, and Weikert, and Petitioner asserts the same general counter-arguments in its Sur-Reply. PO Resp. 56–60; Pet. Reply 26; Sur-Reply 28–29. Accordingly, for the reasons set forth above with respect to the combination of Kurtz, Swinger, and Weikert, Petitioner demonstrates by a preponderance of the evidence that claim 25 would have been obvious over Blumenkranz, Kurtz, and Weikert.

6. *Analysis – Claim 2*

Claim 2 depends from claim 1 and requires “wherein the second treatment pattern is delivered along corneal tissue anterior to the limbus tissue of the patient’s eye.” Ex. 1007, 14:25–27. Because we find that claim 2 would have been obvious over Kurtz, Swinger, and Weikert, we do not address whether claim 2 would have also been obvious over Blumenkranz, Kurtz, and Weikert.

7. *Claim 16 over Blumenkranz, Kurtz, and Weikert, as well as Blumenkranz, Kurtz, Weikert, and Swinger*

Claim 16 depends from claim 1 and further requires “wherein the relaxation incision is formed from the inside of the target tissue towards the outside of the target tissue without extending to the outside of the target tissue.” Ex. 1007, 15:50–53.

Petitioner contends that claim 16 would have been obvious over both Blumenkranz, Kurtz, and Weikert, as well as Blumenkranz, Kurtz, Weikert,

and Swinger. Pet. 40–41, 45–46. Petitioner’s arguments with respect to claim 16 rely on forming incisions in the cornea from a posterior-to-anterior direction. *Id.* As we decline to address whether Blumenkranz would have been modified for surgery on the cornea, and because we have found that claim 16 would have been obvious over Kurtz, Swinger, and Weikert, we do not address Petitioner’s arguments with respect to whether claim 16 would have been obvious based on its asserted Blumenkranz-based grounds.

III. CONCLUSION¹¹

Based on the evidence and arguments presented during the trial, Petitioner has shown by a preponderance of the evidence that claims 1–17 and 20–26 are unpatentable.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–17, 20–26 ¹²	103(a)	Blumenkranz, Kurtz, Weikert	1–15, 17, 20– 26	
16 ¹³	103(a)	Blumenkranz, Kurtz, Weikert, Swinger		

¹¹ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

¹² For the reasons discussed above, we do not address whether claim 16 would have been obvious over Blumenkranz, Kurtz, and Weikert.

¹³ For the reasons discussed above, we do not address Petitioner’s alternative ground addressing whether claim 16 would have been obvious over Blumenkranz, Kurtz, Weikert, and Swinger.

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–3, 6, 16, 17, 20–26	103(a)	Kurtz, Swinger, Weikert	1–3, 6, 16, 17, 20–26	
4, 5, 7– 15	103(a)	Kurtz, Swinger, Weikert, and Benedikt	4, 5, 7–15	
Overall Outcome			1–17, 20–26	

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–17 and 20–26 of the '024 patent are unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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